

notebook

August 27, 2025

1 To do

- Find the alignment problem of 150 ms (motor?) in group 4 (experiment=2AFC_4)
- ~~Repeat all analyses per mouse (figure with a subplots)~~
- Check ILI distributions using only the first interval (between 1st and 2nd licks) instead of the mean
- Use raw distributions instead of KDE when not comparing distributions. We have **a ton** of trials, so we can afford it
- Log scale in y-axis in RT distributions to stretch around the peak
- Check if the 2 peaks in bimodal RT distributions correspond to the first interval (between 1st and 2nd licks) of the ILI. It could be due to preparatory licks (mouse start licking before the lickport is available and when the lickport does come sometimes it register the 1st lick of the train and sometimes the 2nd). If not:
 - Check split in engaged/disengaged to explain bimodality
- ~~Plot lick rate as discrete distributions (it's not possible to do 0.5 licks). As mean lick rate per ILD too for cleaner plot~~
- Check if bimodality is there in groups with
 - Delay (#4-6)
 - Fixed XYZ positions (#6)
 - No evidence levels (# 5-6)

The autoreload extension is already loaded. To reload it, use:

```
%reload_ext autoreload
```

2 Load behavioral data

3 Filter data

211k trials

4 Add lick data

0.16% of premature lick trials (before response window)

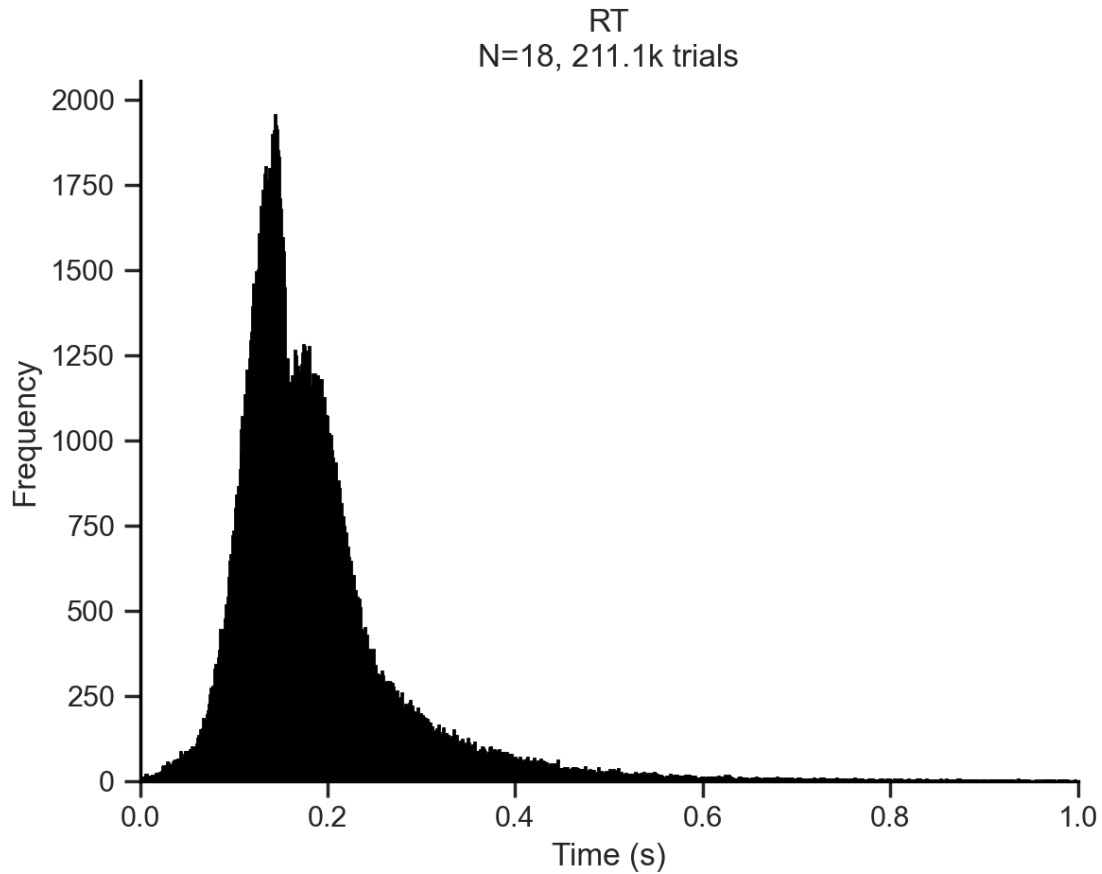
Port1In or Port2In are already lists

```
['absILD', 'LicksLeft', 'LicksRight', 'nLicksLeft', 'nLicksRight', 'nLicks',  
'leftILI', 'rightILI', 'ILI', 'RT', 'RT2', 'SessionHalf']
```

5 Reaction Times (RTs)

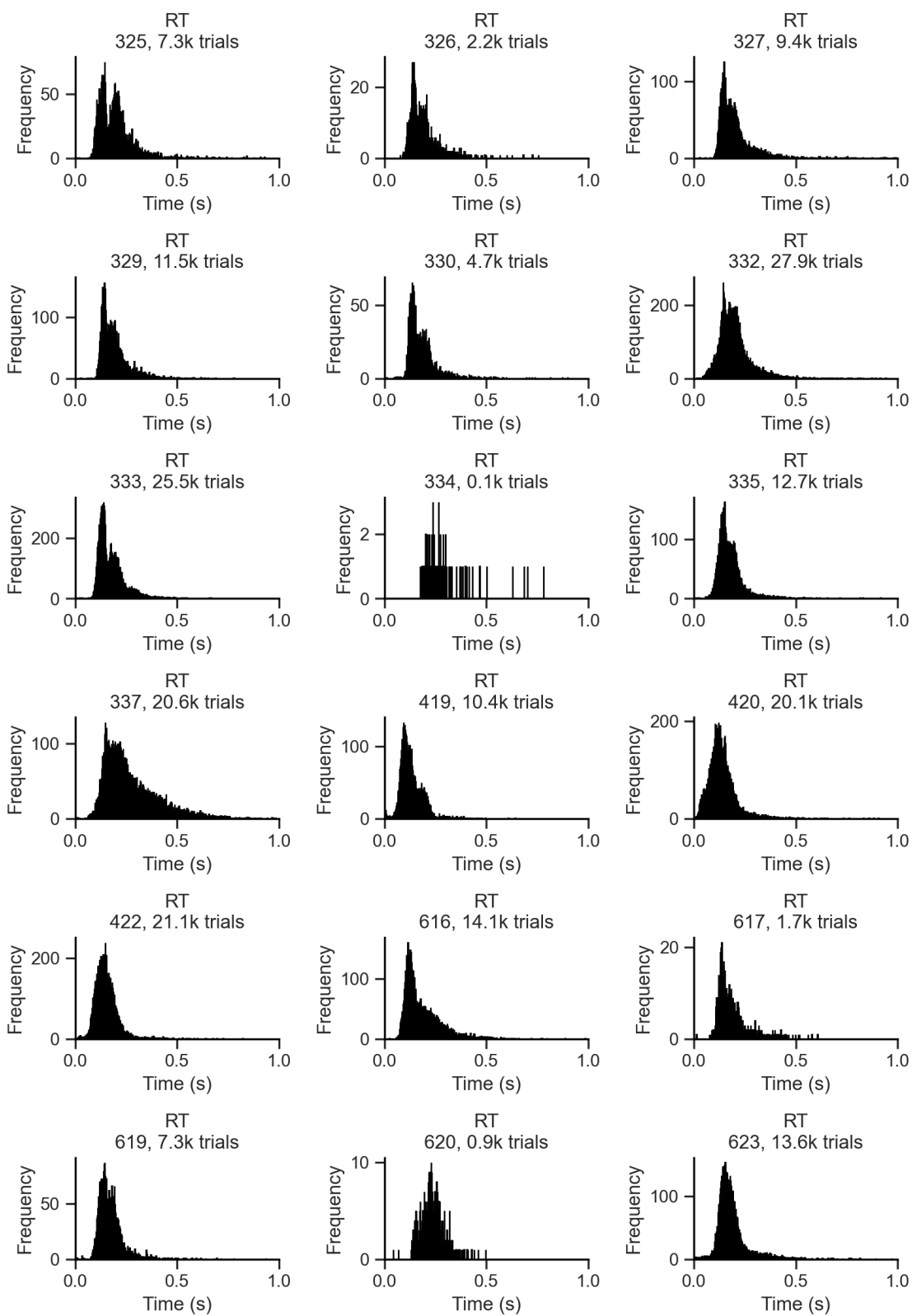
5.1 Distributions

5.1.1 All trials



This is the rawest representation of RT data. It includes all trials from all mice. There is a bimodal distribution of RTs across all mice. The overall distribution has a shoulder-like second peak (lower) after the first one. It is maybe due to the presence of two different strategies in the task: one that involves a fast response and another that involves a slower response. Let's rule out other parsimonious hypotheses first (more likely, easier and faster to check).

First, let's check if this bimodality exist in single subjects, or if in turn it is an averaging artifact.



The bimodality is indeed present in quite some single subjects.

One hypothesis is that the difference evidence levels (ILDs) drives different RTs. To examine this, we can plot a RT distribution per ILD level. To visually compare the distributions, a Kernel Density Estimation (KDE) plot is preferred.

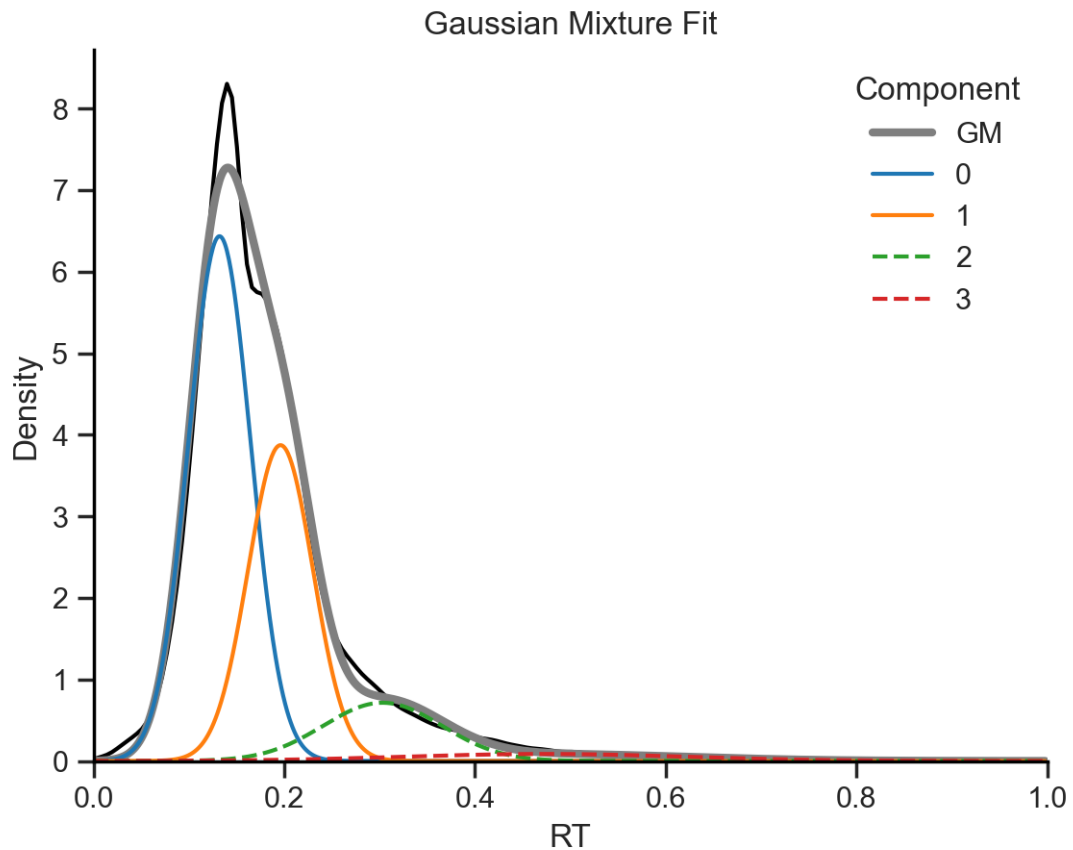
Gaussian Mixture fit

Component 0: mean=0.13 s, std=0.03 s, weight=0.53

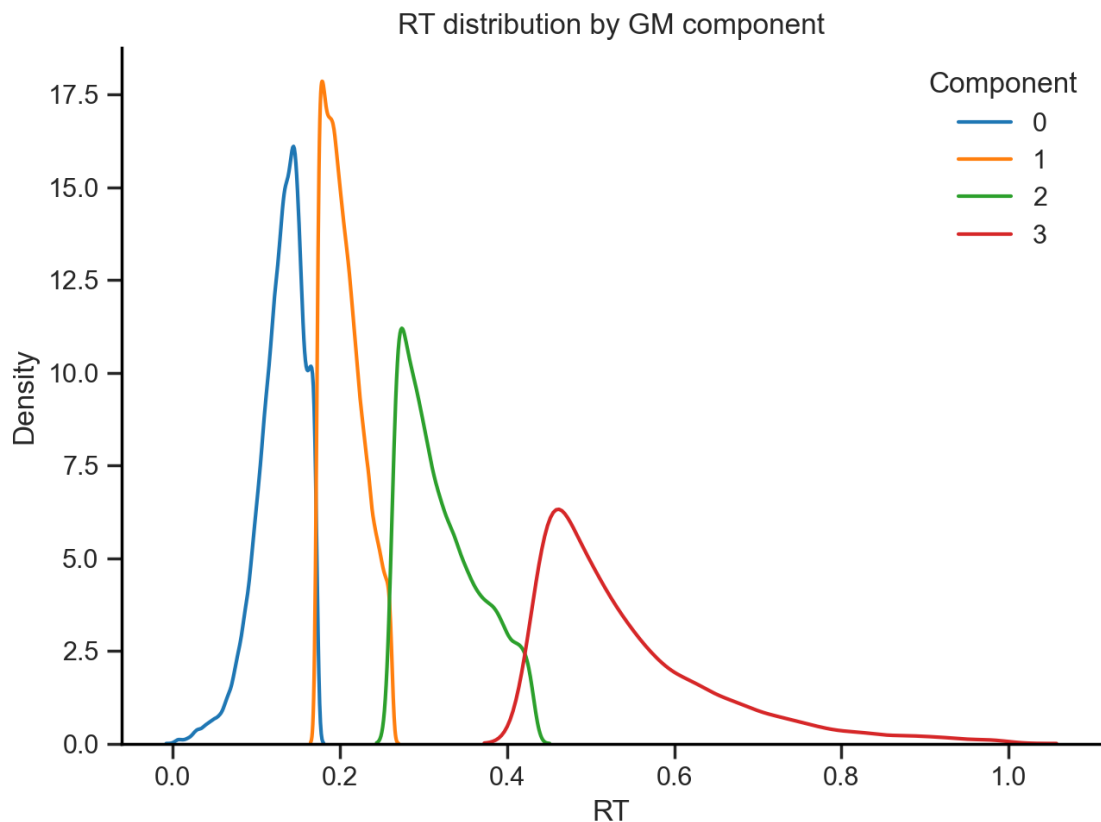
Component 1: mean=0.20 s, std=0.03 s, weight=0.33

Component 2: mean=0.30 s, std=0.06 s, weight=0.11

Component 3: mean=0.47 s, std=0.15 s, weight=0.03



Plot RT distributions grouped by component



Screening of variables to explain the difference between the first 2 components (0 and 1)

```
Index(['Trial', 'Side', 'RepTrial', 'Reward', 'Punish', 'Miss', 'WrongLick',
      'Hit', 'AfterHit', 'Choice', 'RepChoice', 'Response', 'TrialStart',
      'TrialEnd', 'TriallLen', 'StimStart', 'StimEnd', 'StimLen',
      'RespWinStart', 'RespWinEnd', 'RespWinLen', 'Filename', 'Filename2',
      'FilesMatch', 'Message', 'MessageFound', 'SoundLeft', 'SoundRight',
      'Sound', 'ILD', 'absILD', 'ILDRep', 'Port1In', 'Port1Out', 'Port2In',
      'Port2Out', 'ValveLeft', 'ValveRight', 'AW', 'AWTrials', 'Switch',
      'Timeout', 'Fixation', 'Stage', 'Motor', 'REC', 'Progression', 'CB',
      'RespWin', 'ITI', 'WarmUp', 'RecoveryMode', 'P', 'PRight', 'Blocks',
      'BlockLen', 'Task', 'Ephys', 'StimDur', 'Delay', 'VarDelay',
      'SerialPort', 'Protocol', 'Creator', 'Project', 'Experiment', 'Board',
      'Setup', 'NetPort', 'Subject', 'BpodApiVersion', 'Session', 'Date',
      'SessionStart', 'SessionEnd', 'LicksLeft', 'LicksRight', 'nLicksLeft',
      'nLicksRight', 'nLicks', 'leftILI', 'rightILI', 'ILI', 'RT', 'RT2',
      'SessionHalf', 'RTlabelsGM'],
      dtype='object')
```

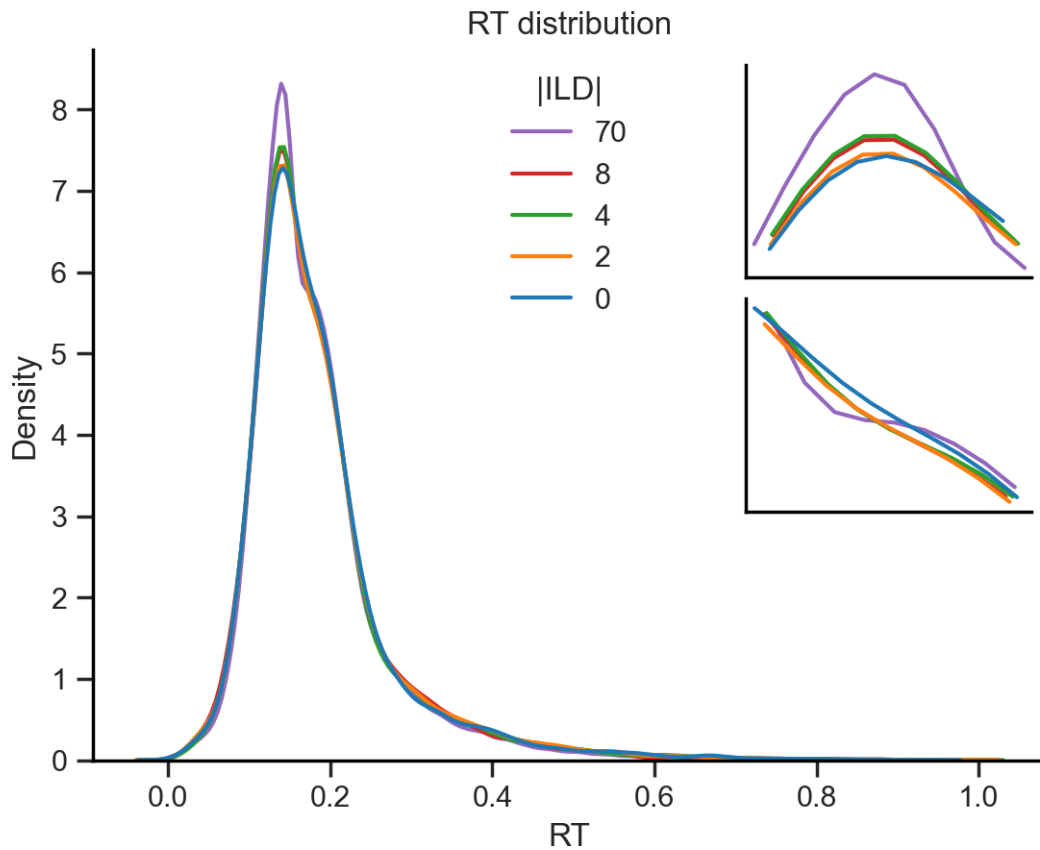
Variable	Type	MeanDiff	pval	pval_adj
----------	------	----------	------	----------

13	SessionHalf	numeric	-0.062407	1.264823e-148	1.770752e-147
11	nLicksRight	numeric	0.398896	2.241887e-110	3.138642e-109
10	nLicksLeft	numeric	0.294755	3.391199e-70	4.747679e-69
4	AfterHit	numeric	0.024331	1.430650e-38	2.002910e-37
3	Hit	numeric	0.023231	7.526695e-35	1.053737e-33
0	Trial	numeric	-8.373798	3.987057e-31	5.581880e-30
9	P	numeric	0.008394	1.433574e-25	2.007003e-24
7	ILD	numeric	2.410915	2.060798e-16	2.885117e-15
5	Choice	numeric	0.018785	5.445209e-15	7.623292e-14
1	Side	numeric	0.016598	5.151238e-12	7.211733e-11
8	ILDRep	numeric	1.878037	1.944585e-10	2.722418e-09
2	RepTrial	numeric	0.014171	4.733627e-09	6.627078e-08
6	RepChoice	numeric	0.008567	3.200018e-04	4.480025e-03
12	absILD	numeric	0.101399	4.564564e-01	6.390390e+00

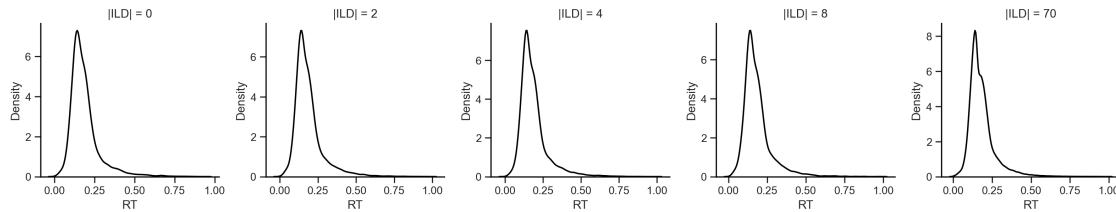
5.1.2 Splits

ILD

Mean RT = 0.142 s



Too many lines can make on top of each other make it hard to disentangle the ILD impact on RT, so we can plot the RT distribution for each ILD level separately.



Although the bimodality is present in all ILD levels, the second peak is more pronounced for lower ILD levels. This suggests that the bimodality is modulated by evidence.

Mean RT = 0.136 s
Mean RT = 0.151 s
Mean RT = 0.150 s
Mean RT = 0.162 s
Mean RT = 0.142 s
Mean RT = 0.174 s
Mean RT = 0.135 s

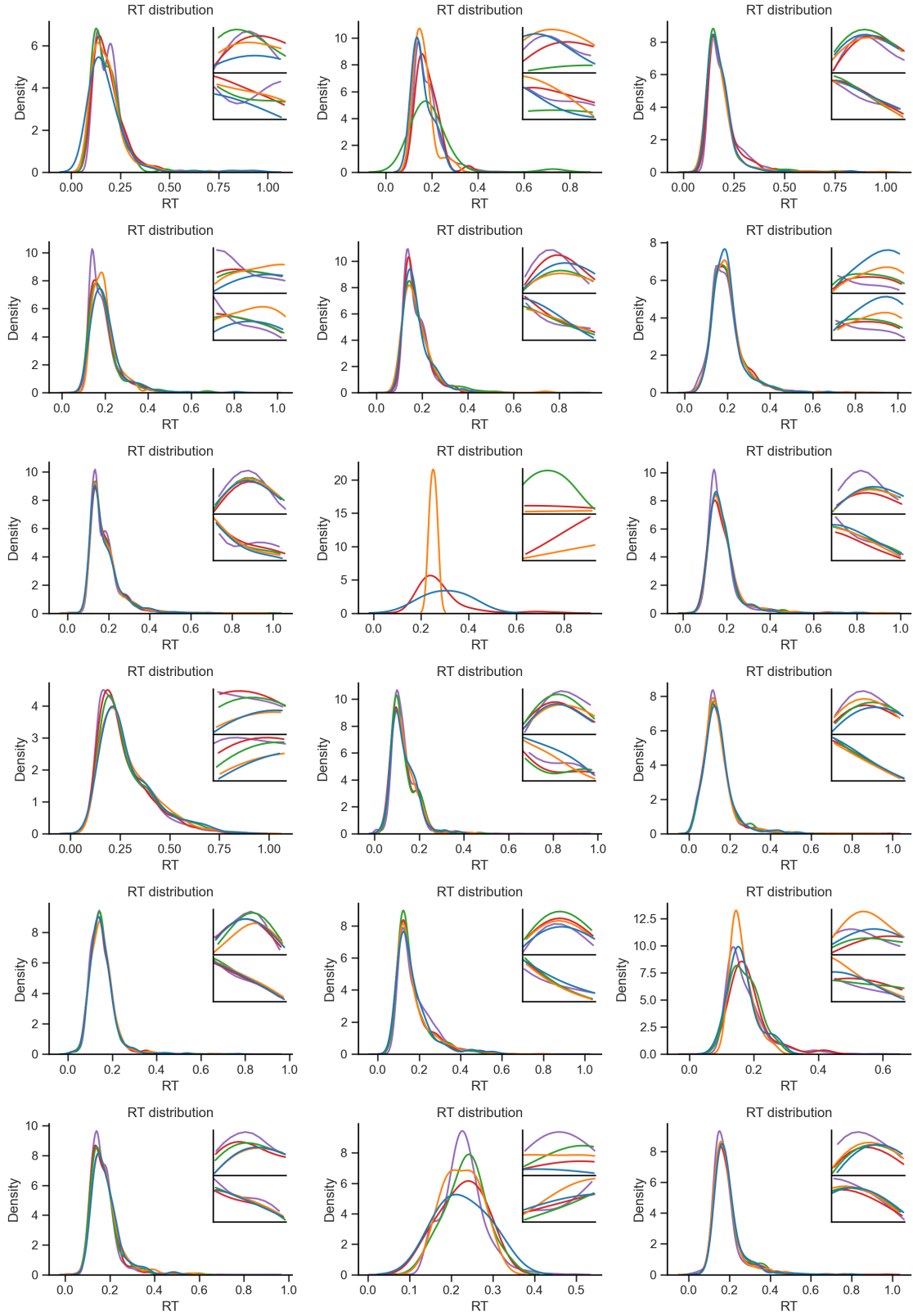
C:\Users\Usuario\PycharmProjects\licks\licks.py:399: UserWarning: Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this warning.

```
line = plt.gca().lines[-1]
```

Mean RT = 0.259 s
Mean RT = 0.148 s
Mean RT = 0.194 s
Mean RT = 0.099 s
Mean RT = 0.121 s
Mean RT = 0.140 s
Mean RT = 0.122 s
Mean RT = 0.148 s
Mean RT = 0.143 s
Mean RT = 0.226 s
Mean RT = 0.158 s

C:\Users\Usuario\PycharmProjects\licks\licks.py:514: UserWarning: This figure includes Axes that are not compatible with tight_layout, so results might be incorrect.

```
"""
```

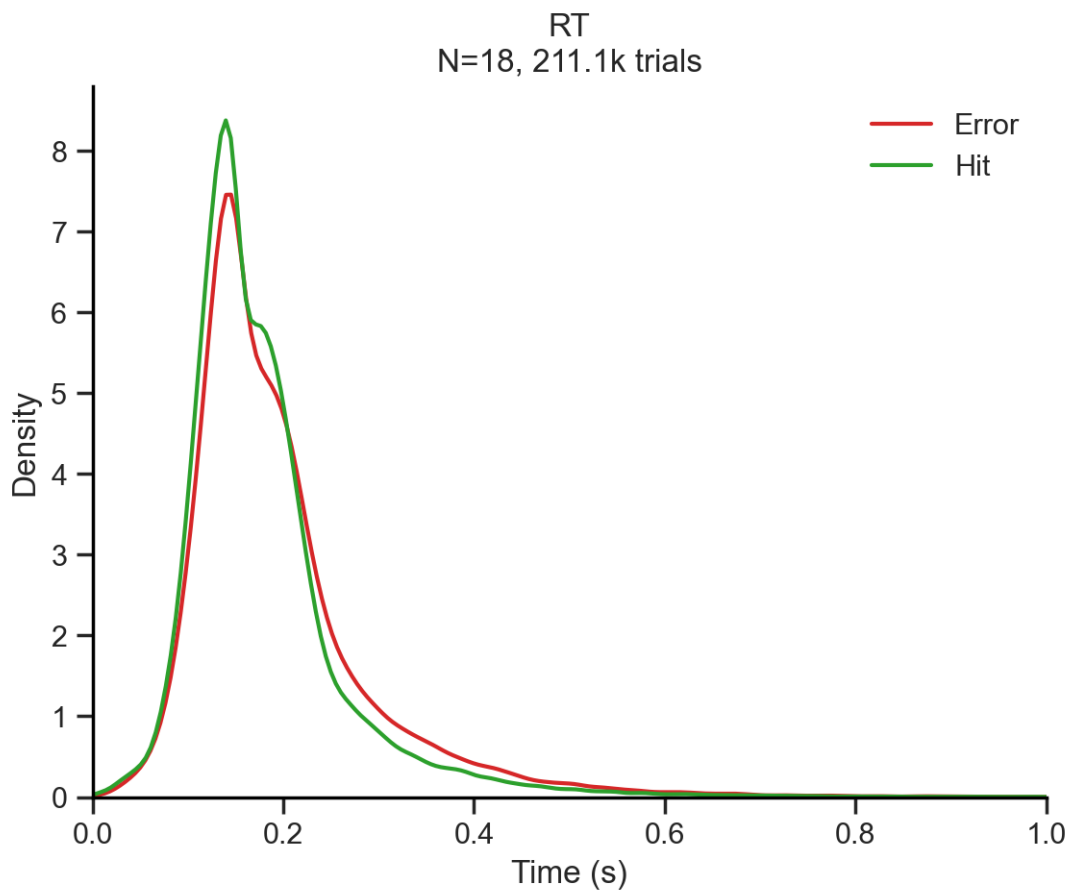


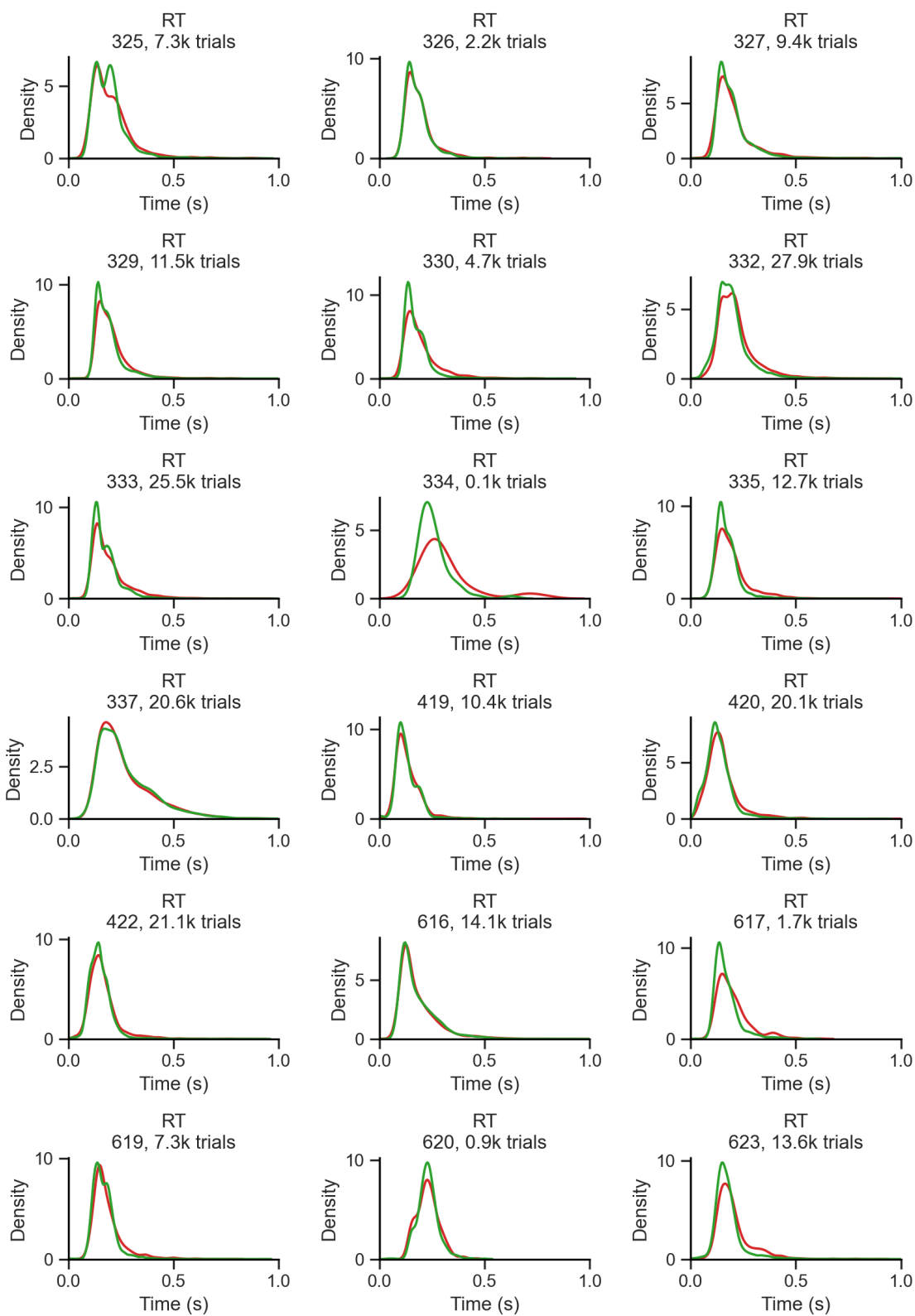
The ILD alone does not seem to explain the **existence** of bimodality in the RT distribution, as the bimodality is present across all ILD levels (for subjects that did show bimodality in their RTs).

Nevertheless, zooming in the peak of the distribution (0.1-0.2 s, first inset), we can see that the RTs are faster for lower ILD levels, which is expected as the evidence is more certain. What is surprising is for this modulation to show up despite the fact that mice were head-fixed and the stimulus was fixed duration. In addition, zooming in the second peak of the distribution (0.15-0.2 s, second inset), we can see that the second peak is more pronounced for lower ILD levels, suggesting that the bimodality is actually modulated by ILD level.

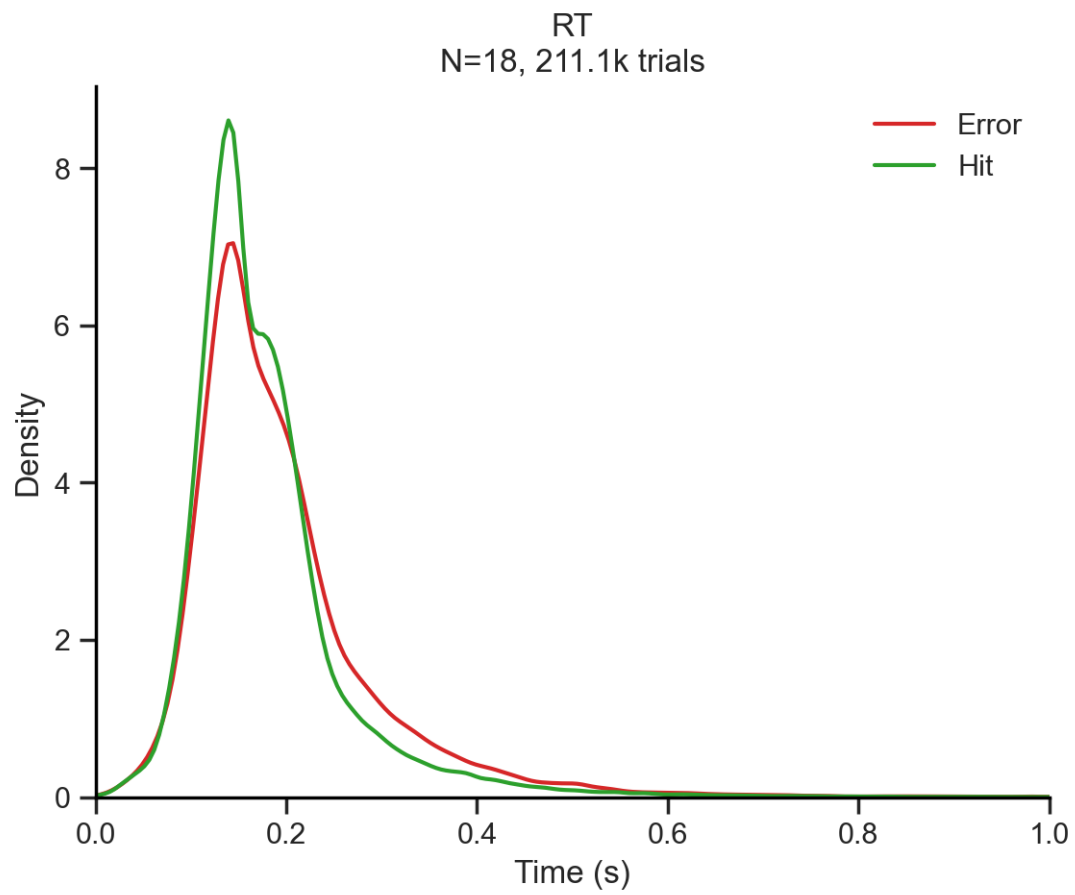
Another hypothesis is that the bimodality is driven by differences in behavioral variables such as outcome, previous outcome, choice, previous choice, repeat trial, stimulus, or session half. To examine this, we can plot a RT distribution per behavioral variable split in 2 conditions.

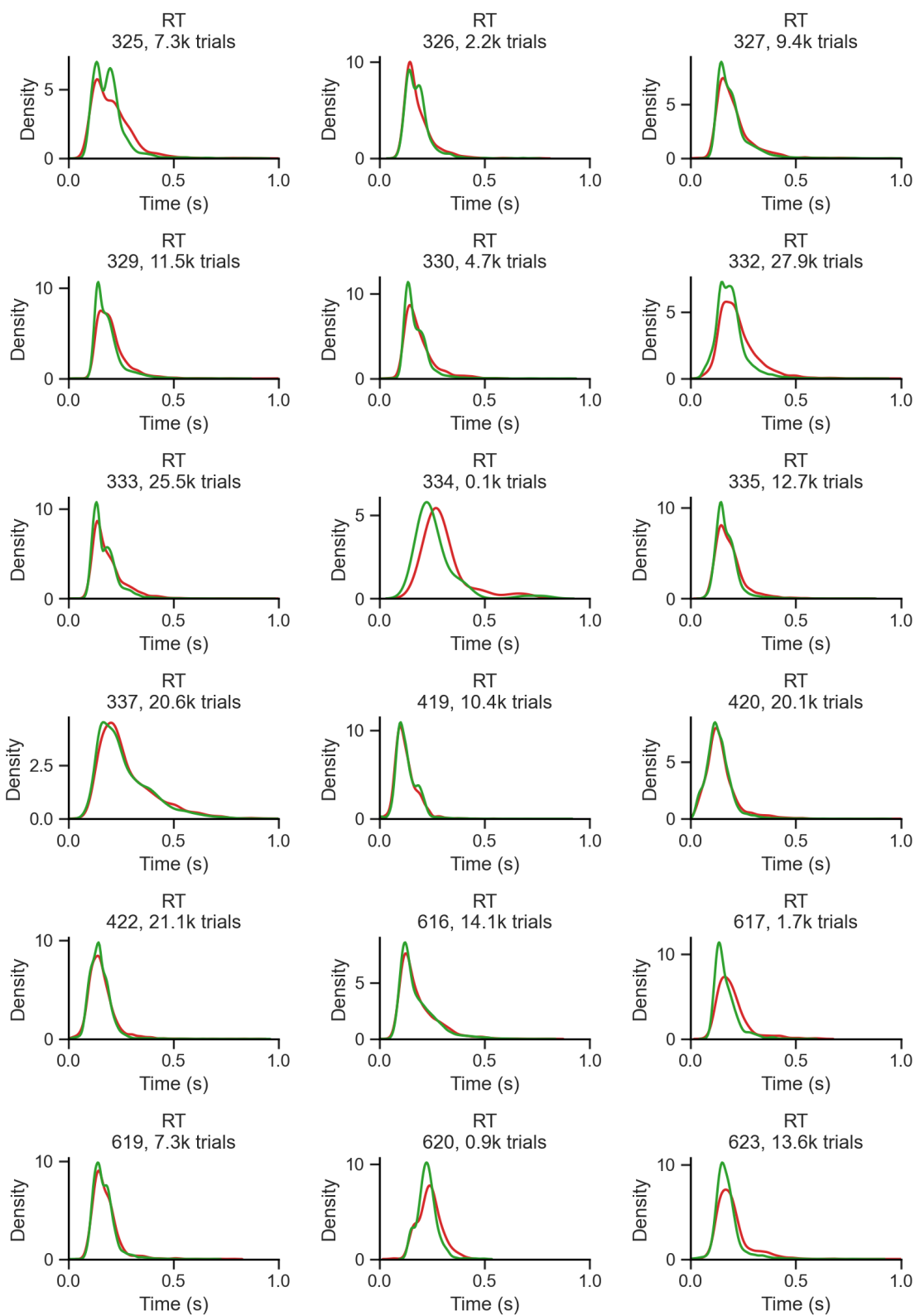
Outcome



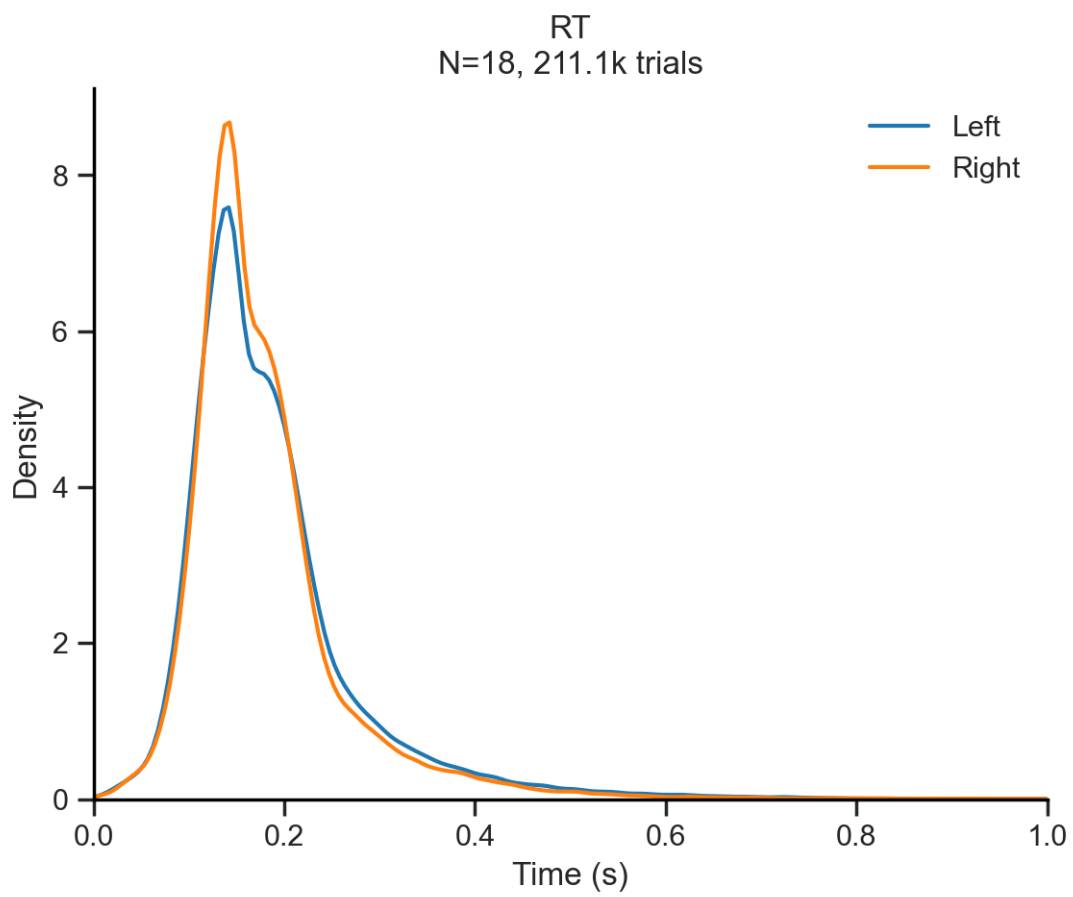


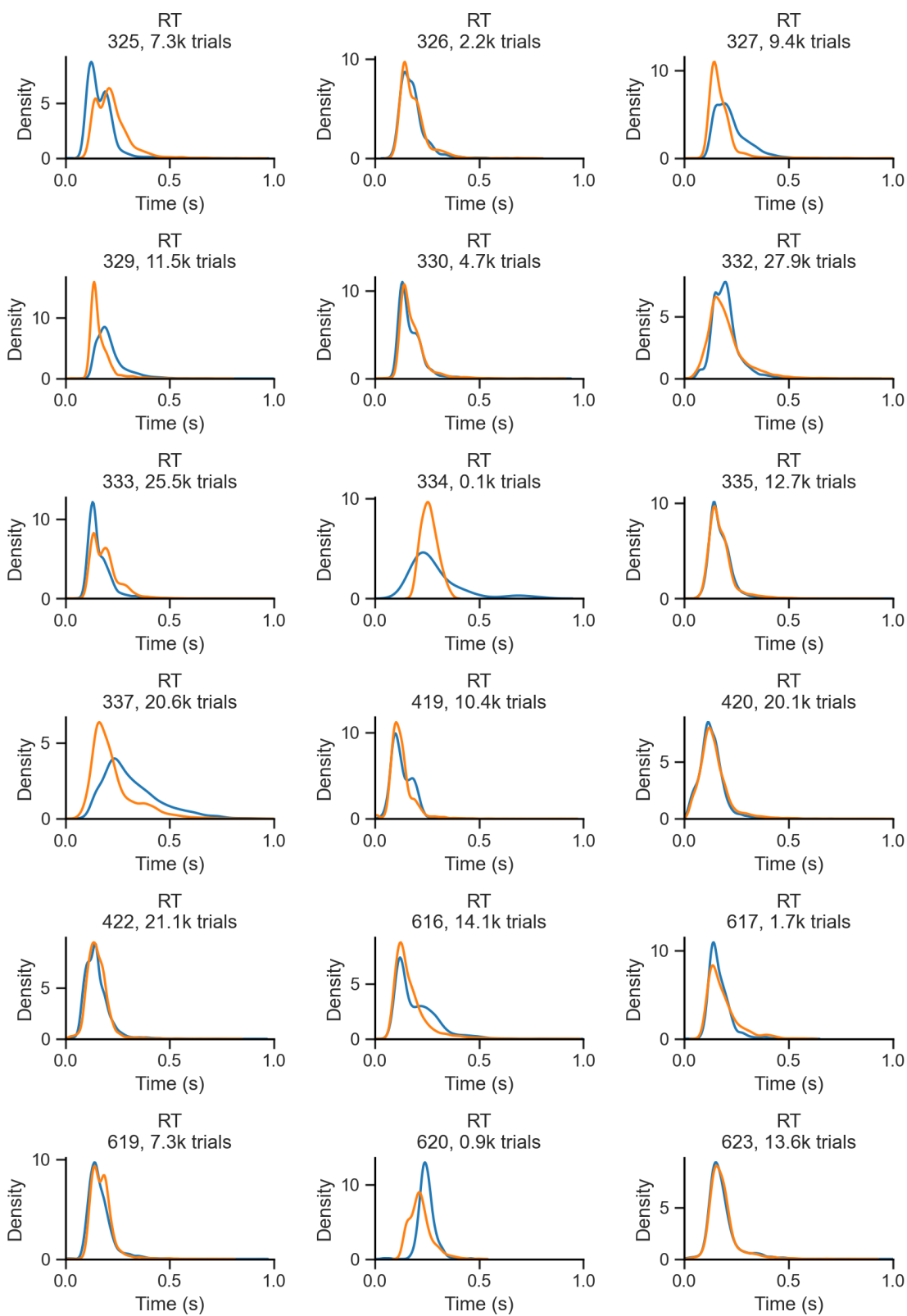
Previous outcome



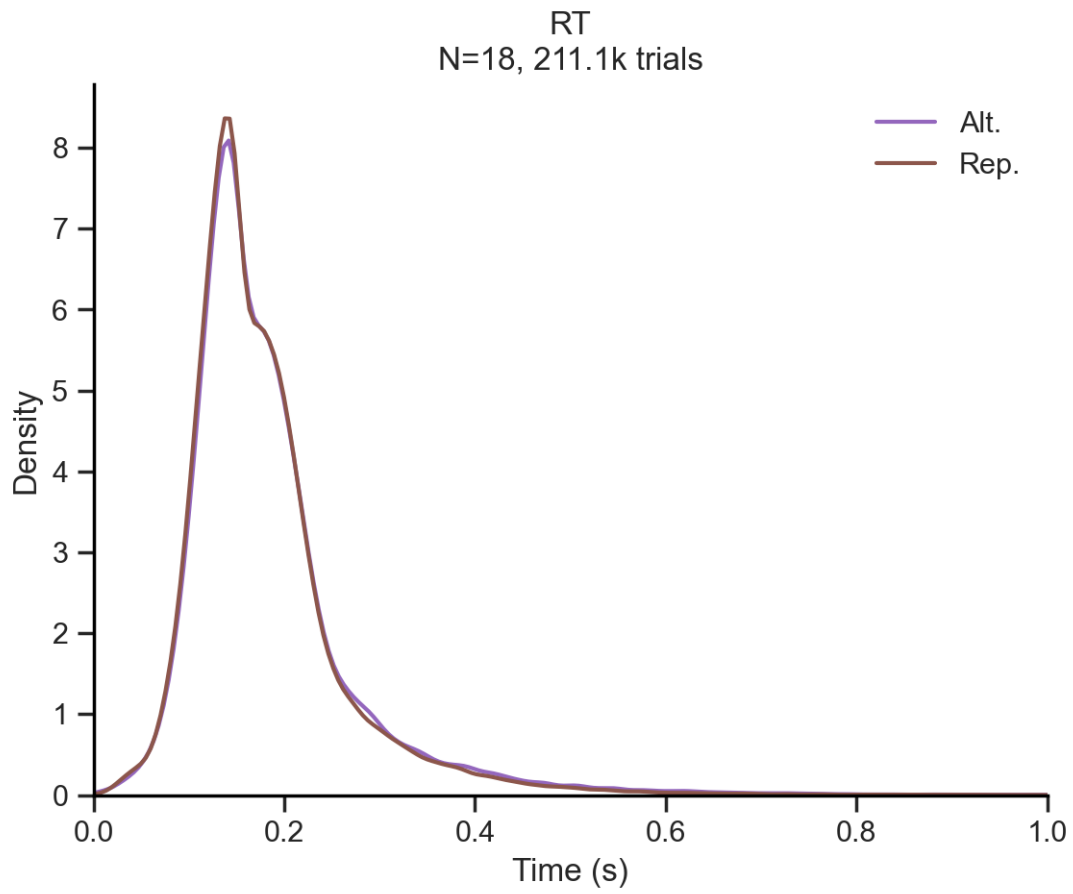


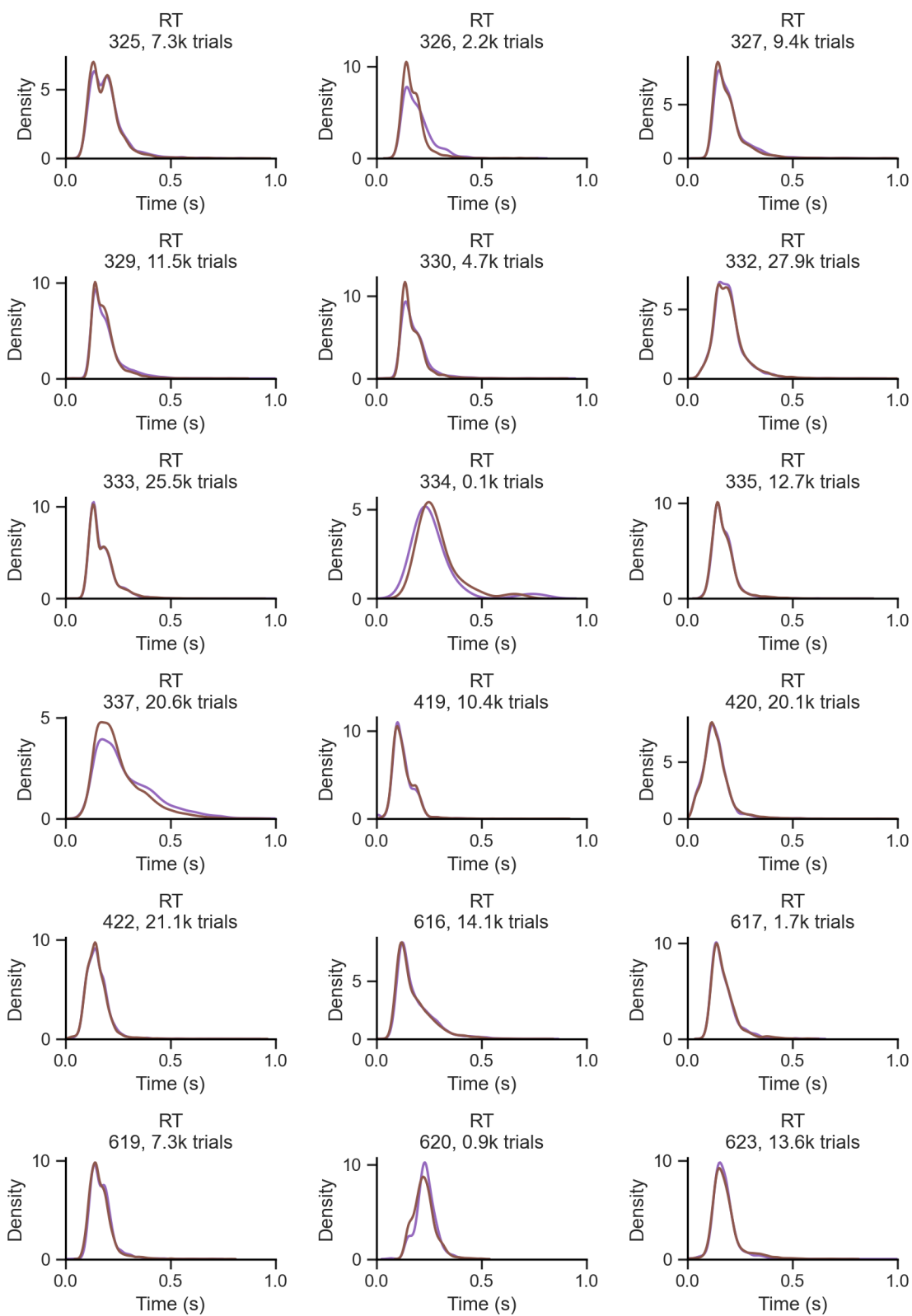
Choice



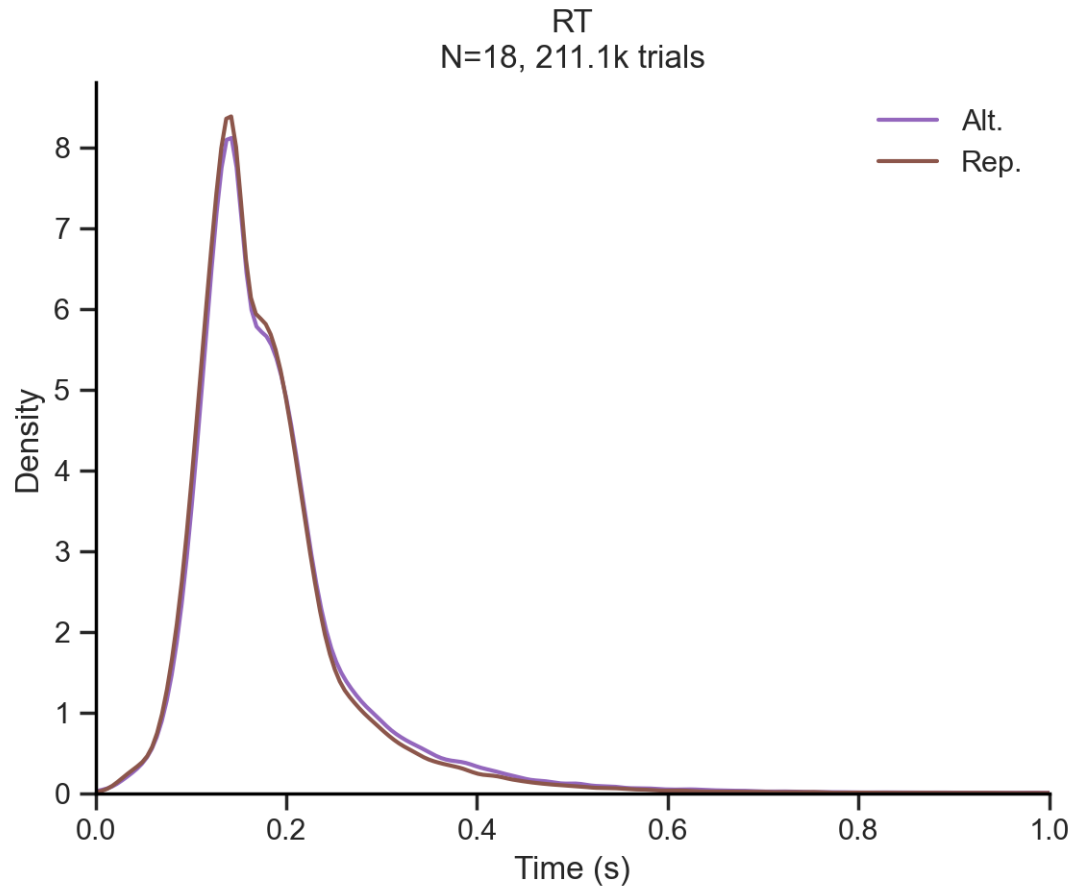


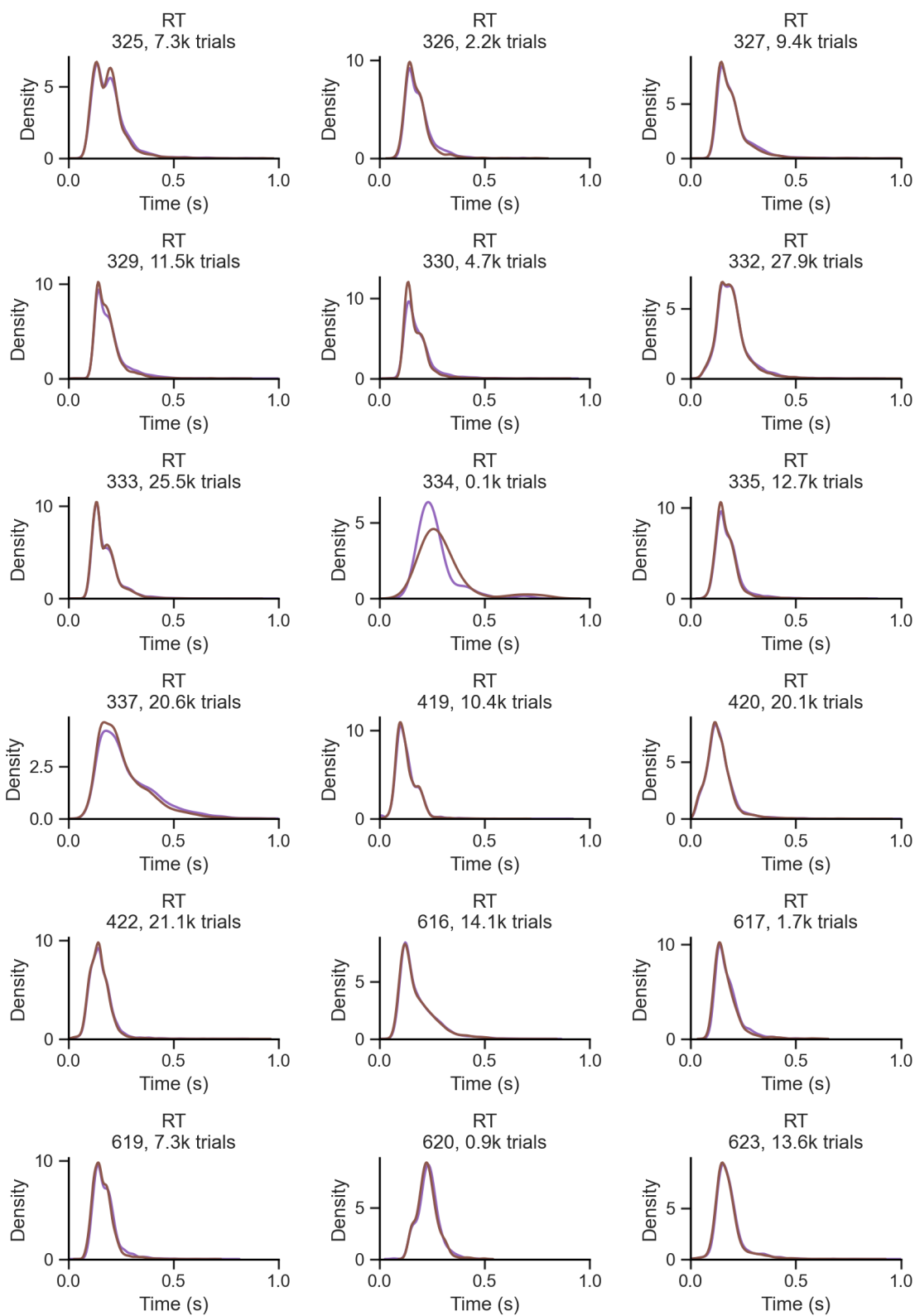
Repeat choice



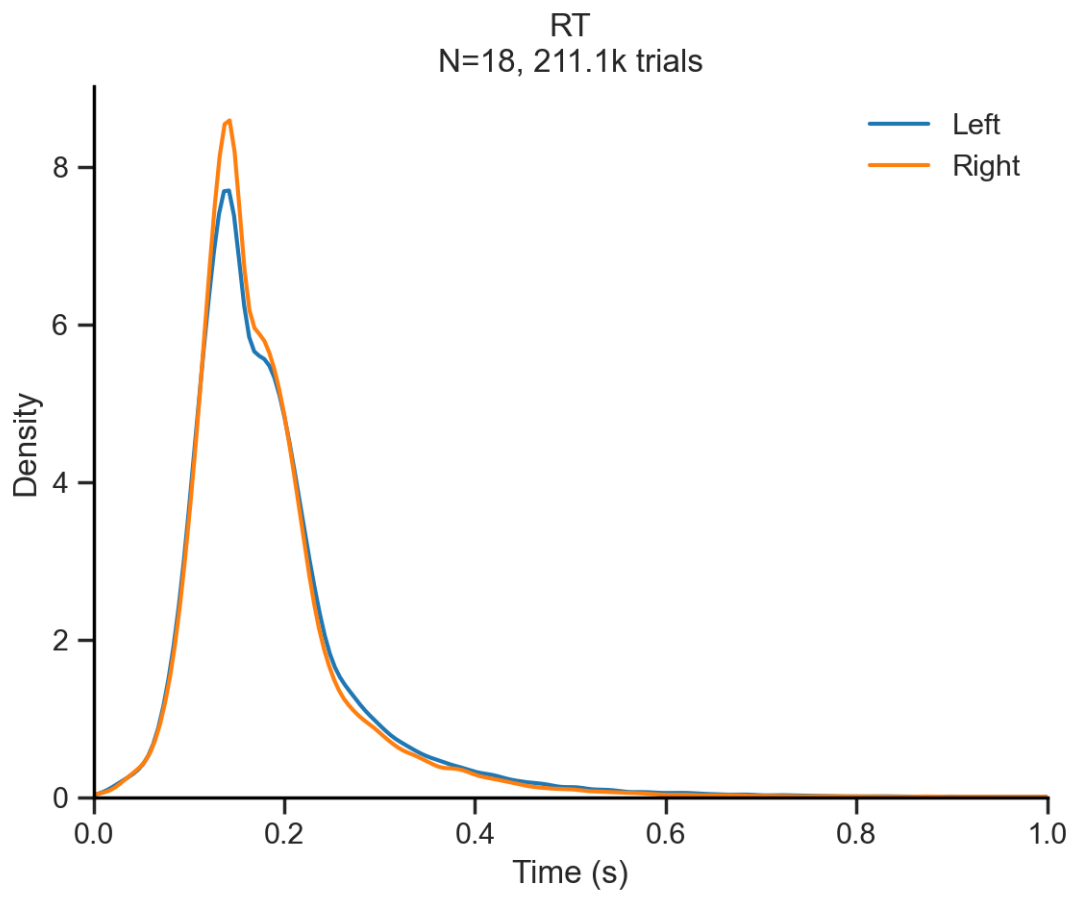


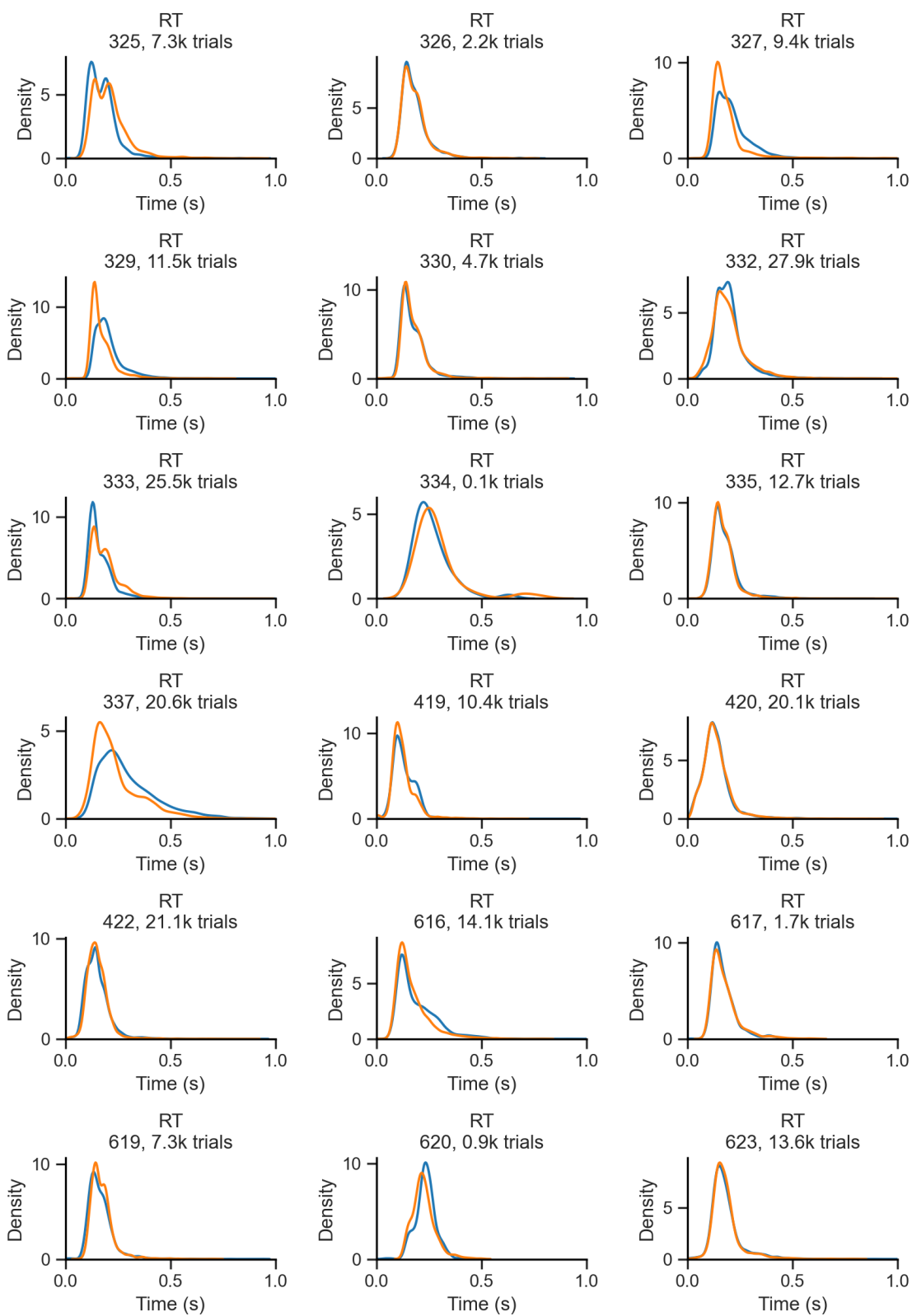
Repeat trial Trials in which the stimulus (rewarded) side coincides with the previous choice





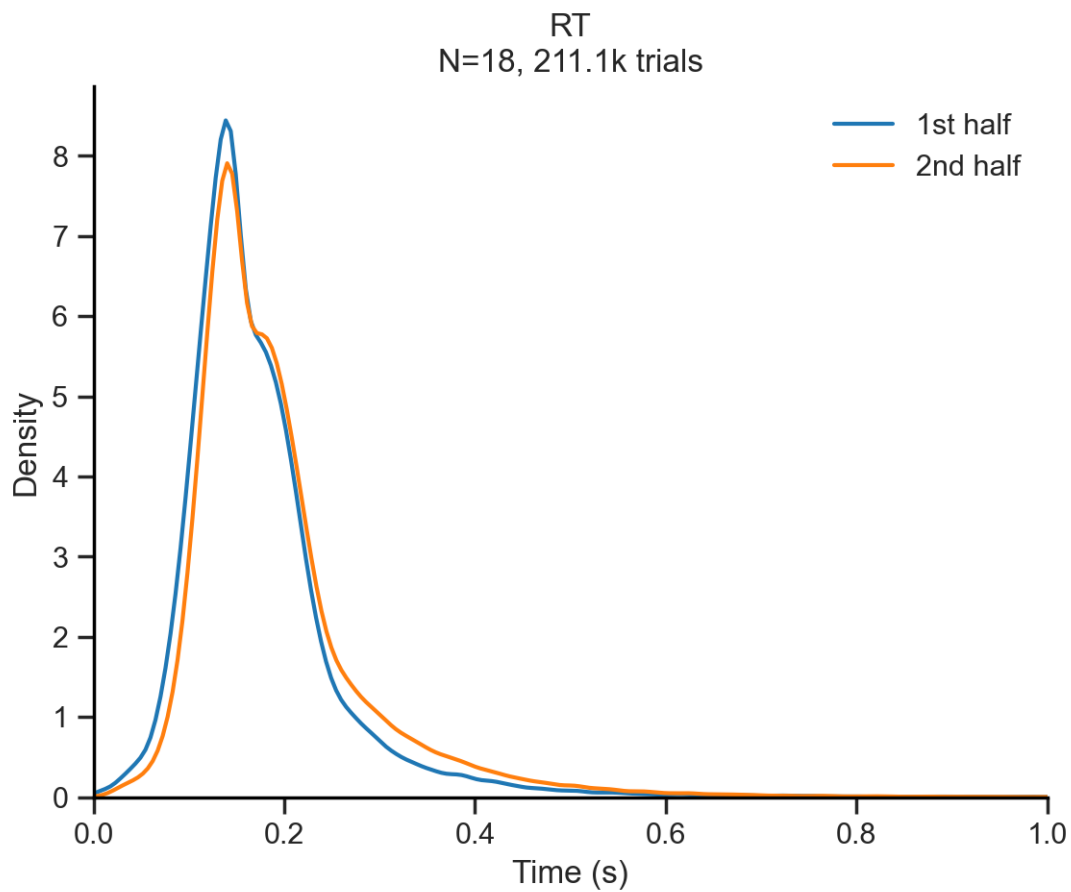
Stimulus

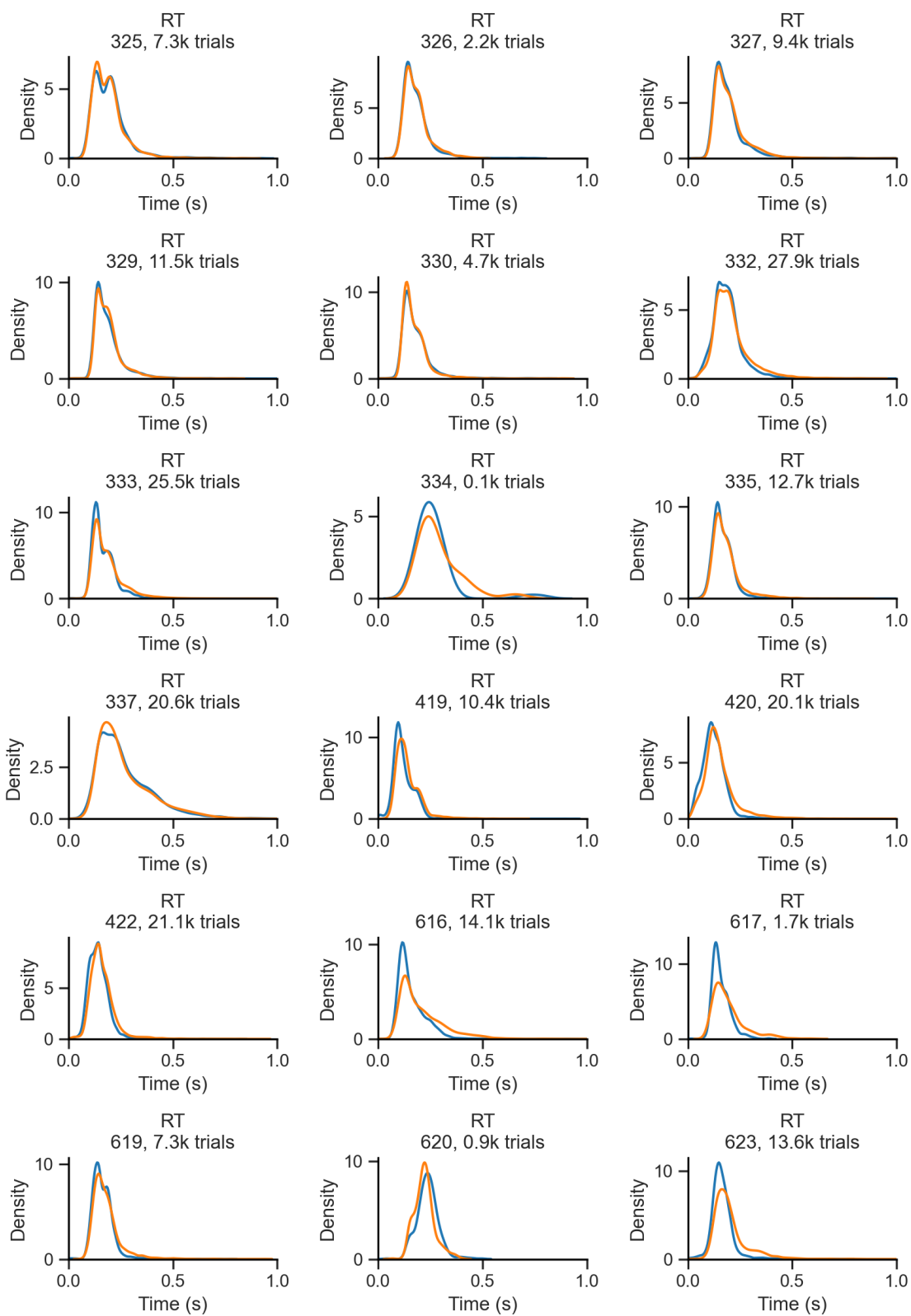




This bimodality can not be explained by differences in behavioral variables such as outcome, choice, stimulus, repeat, or previous outcome.

1st vs 2nd session half





The bimodality is present in both halves of the session, suggesting that it is not driven by a change in the internal state of the animal. The RT distribution is similar in both halves of the session, with a slight shift towards slower RTs in the second half.

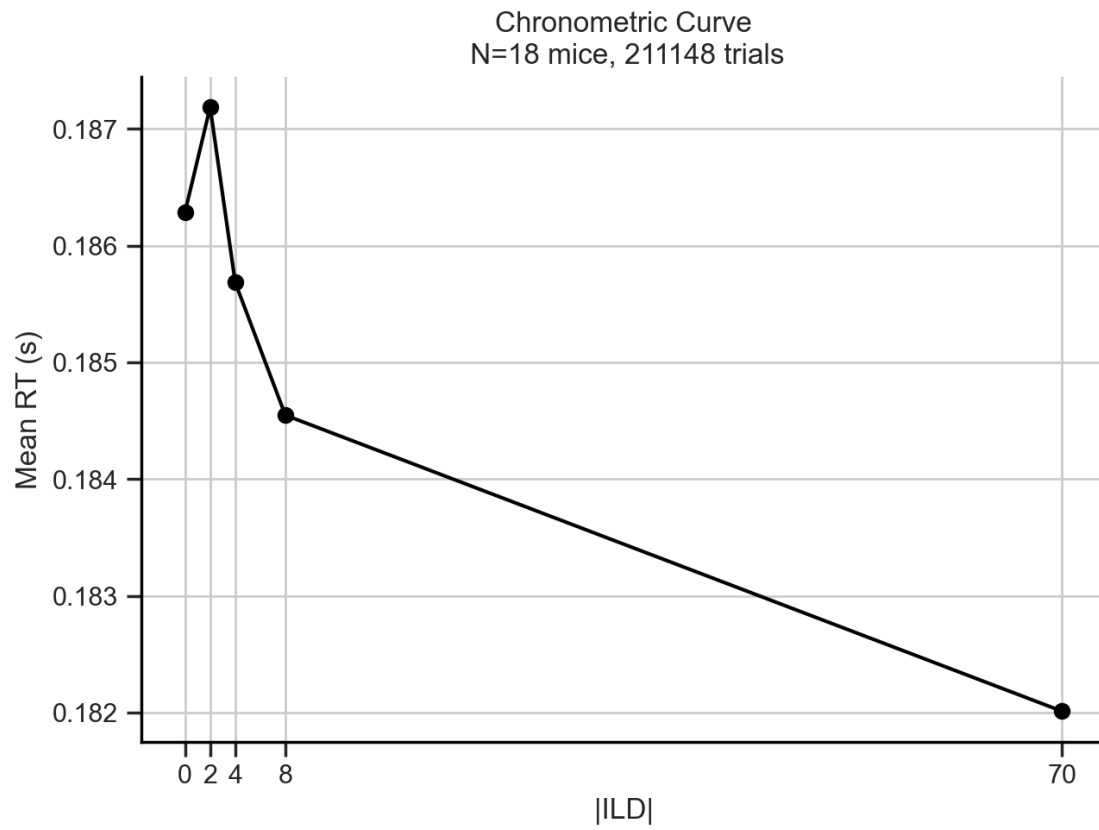
Another possibility is that the bimodality just results from the individual variability in the RTs of each subject. To examine this, we can plot the RT distribution for each subject separately. If the bimodality is driven by individual variability, we would expect to see a unimodal distribution in each subject.

Hidden Markov Model (HMM) Another possibility is that the bimodality reflects the a change in the internal state of the animal (e.g. motivation, reduced by satiation as time passes) that is not captured by the behavioral variables. To examine this, we can split the data of each session into first and second half and plot the RT distribution for each half. If the bimodality is driven by a change in the internal state of the animal, we would expect to see a difference in the RT distribution between the first and second half of the session.

5.2 Chronometric Curves

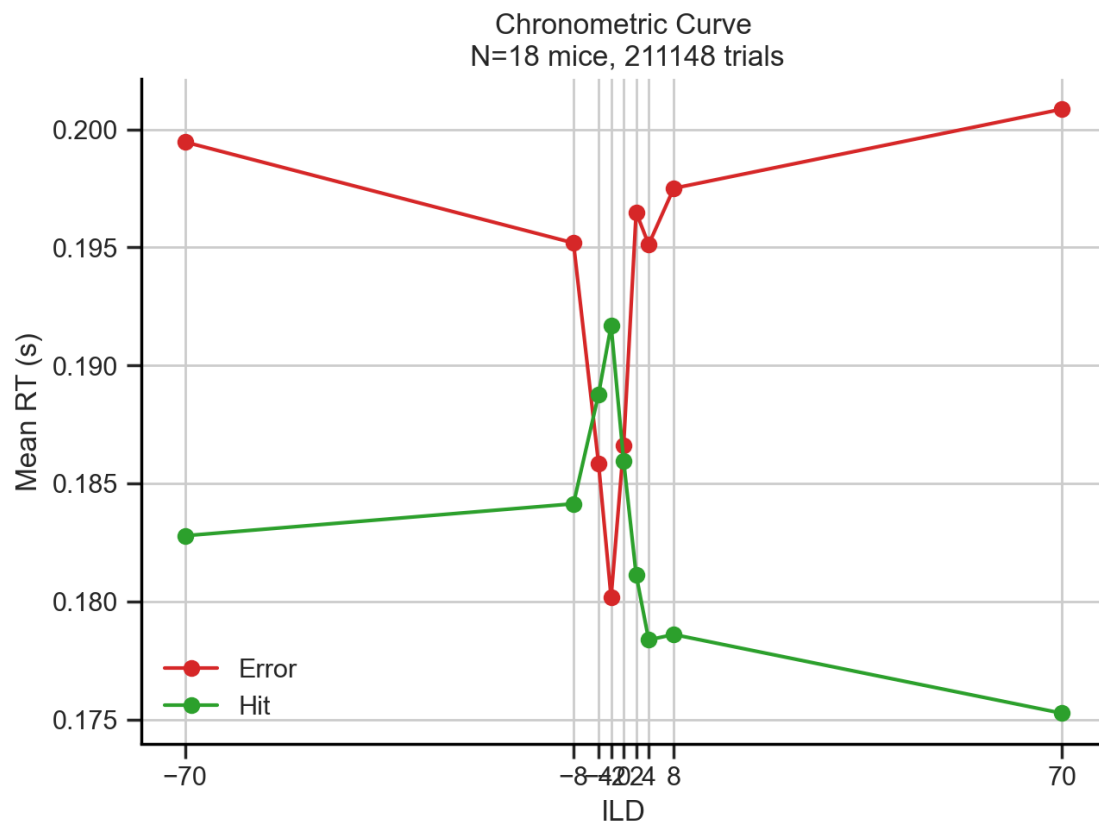
5.2.1 All trials

	absILD	RT
0	0.0	0.186289
1	2.0	0.187186
2	4.0	0.185685
3	8.0	0.184547
4	70.0	0.182010

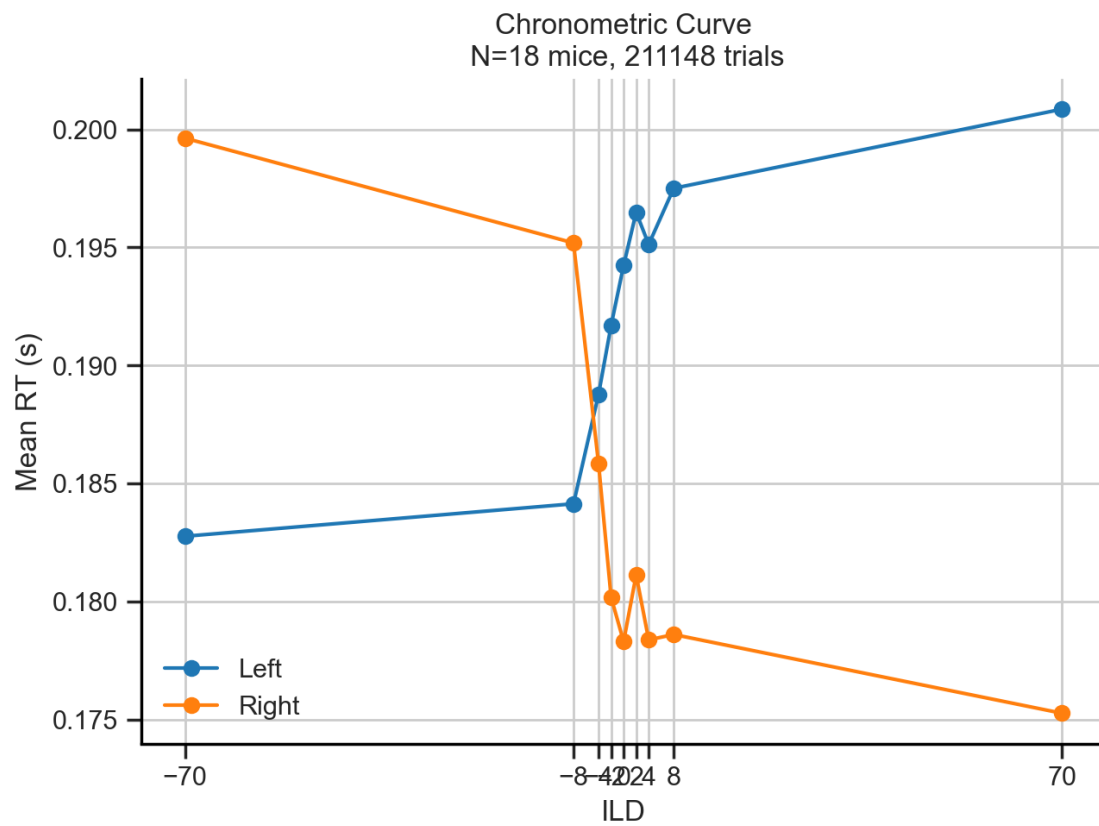


5.2.2 Splits

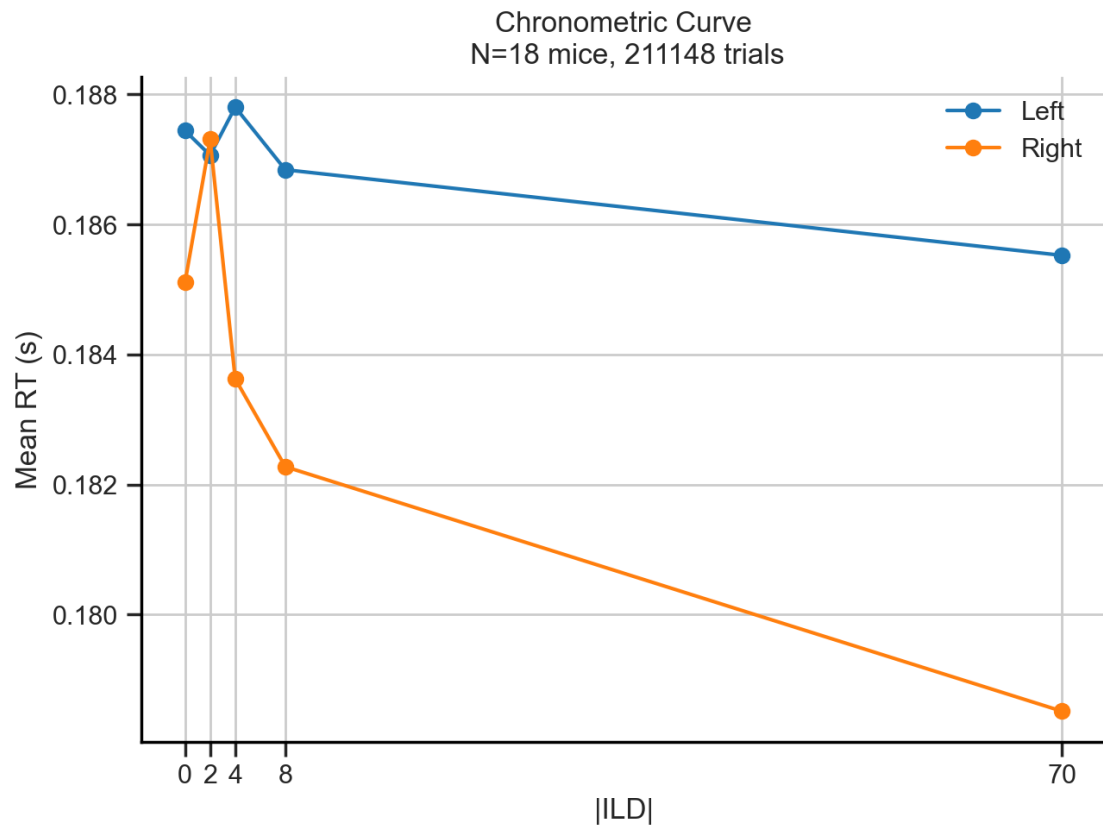
Outcome



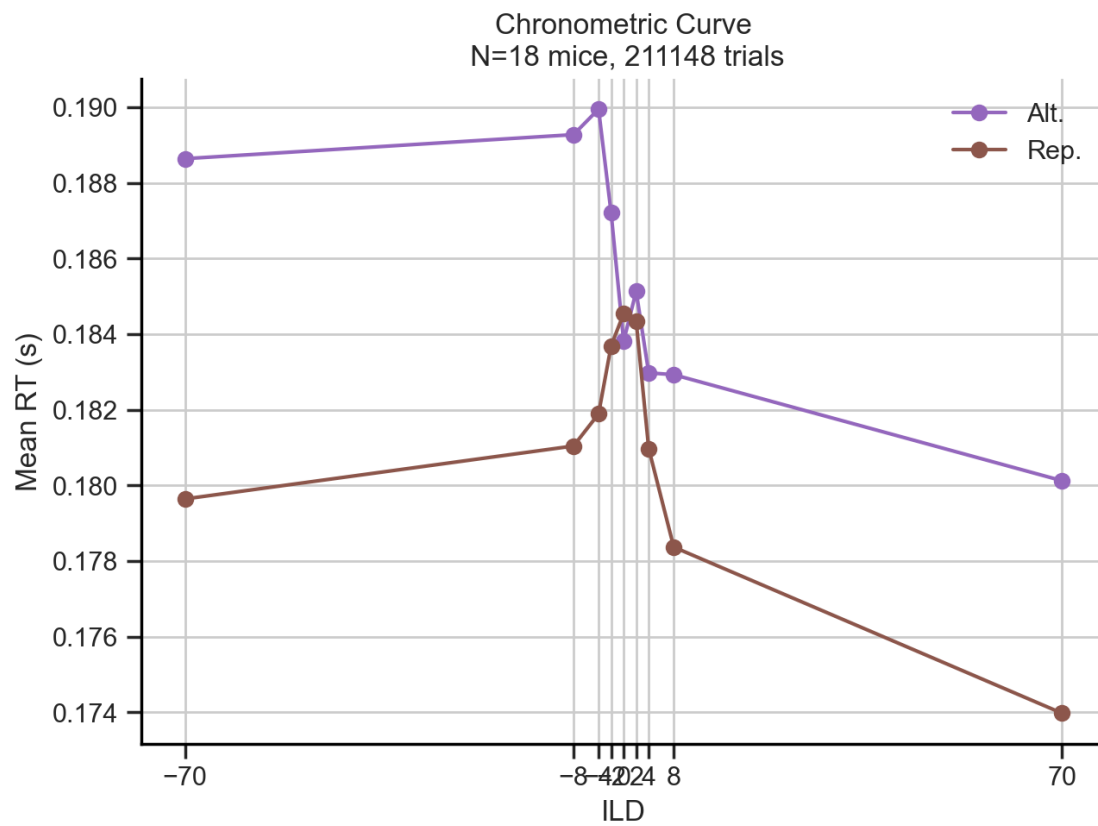
Choice



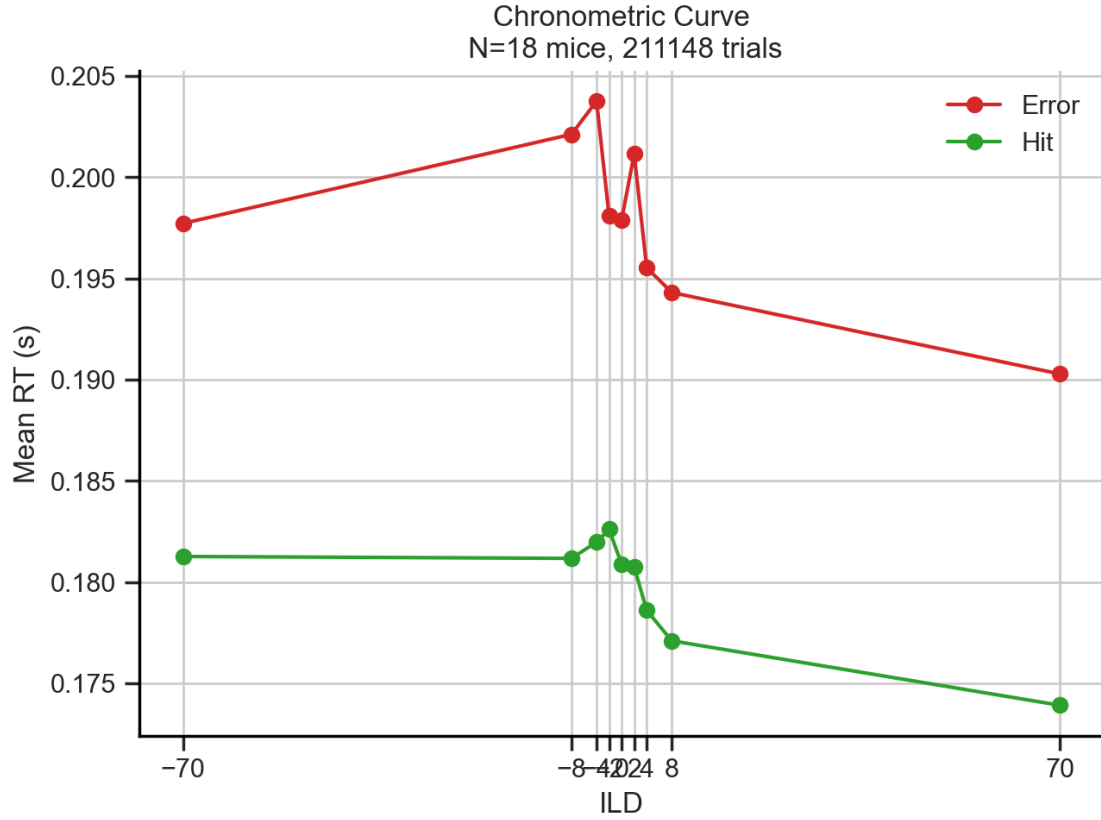
Stimulus



Repeat



Previous outcome



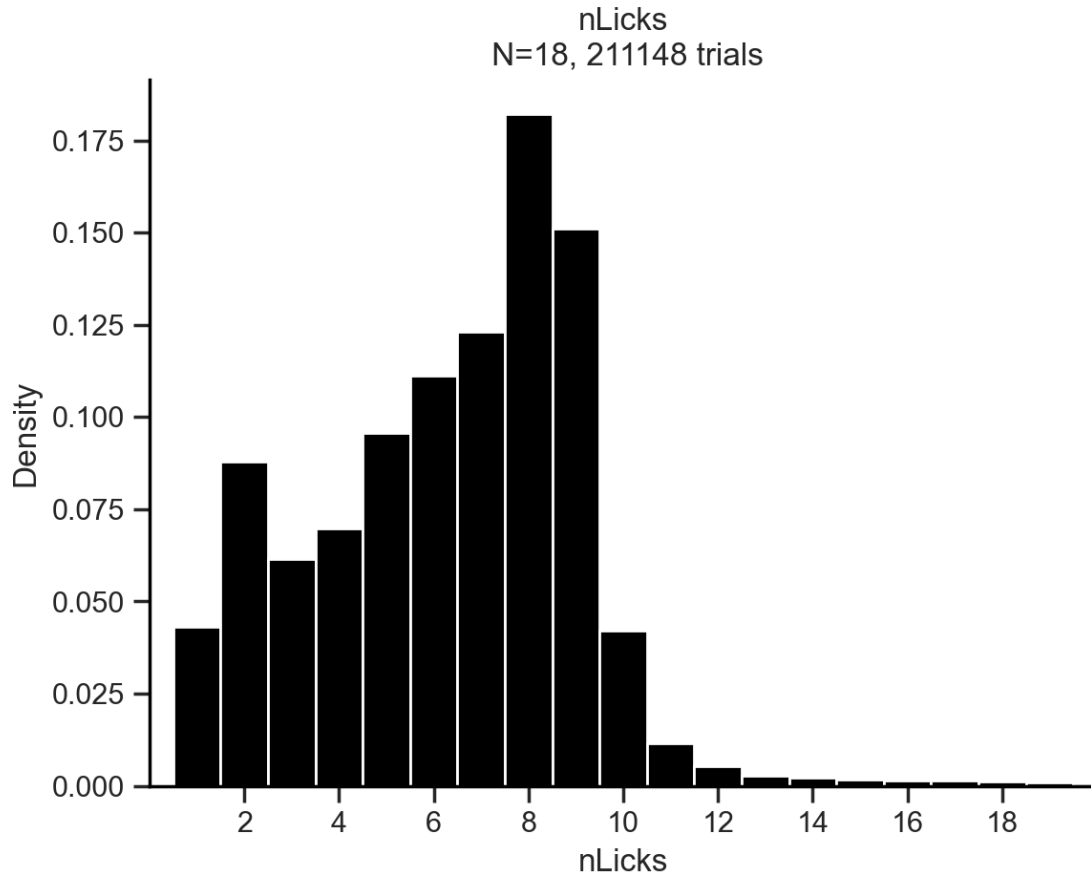
6 Lick Rate

As the number of licks depends on the time the lickport was available (motor in), which ultimately depends on the outcome (correct vs error), a time window to count the licks needs to be established. As the time to collect water in correct (rewarded) trials was always 1 s, and the timeout period following error trials varied from 2-5 s, the chosen time window to count the number of licks was 1 s. This is very convenient for the lick rate, as it is defined as

$$\text{Lick rate} = \frac{N_{\text{licks}}}{\text{time (s)}}$$

so that $N \text{ licks} = \text{Lick Rate}$

6.1 All trials



We can observe a first peak at 2 licks (error trials), and an sawtooth increasing second peak with max at 8 licks (correct trials). Let's split the data into several variables (like with RT) to observe how they modulate the lick rate.

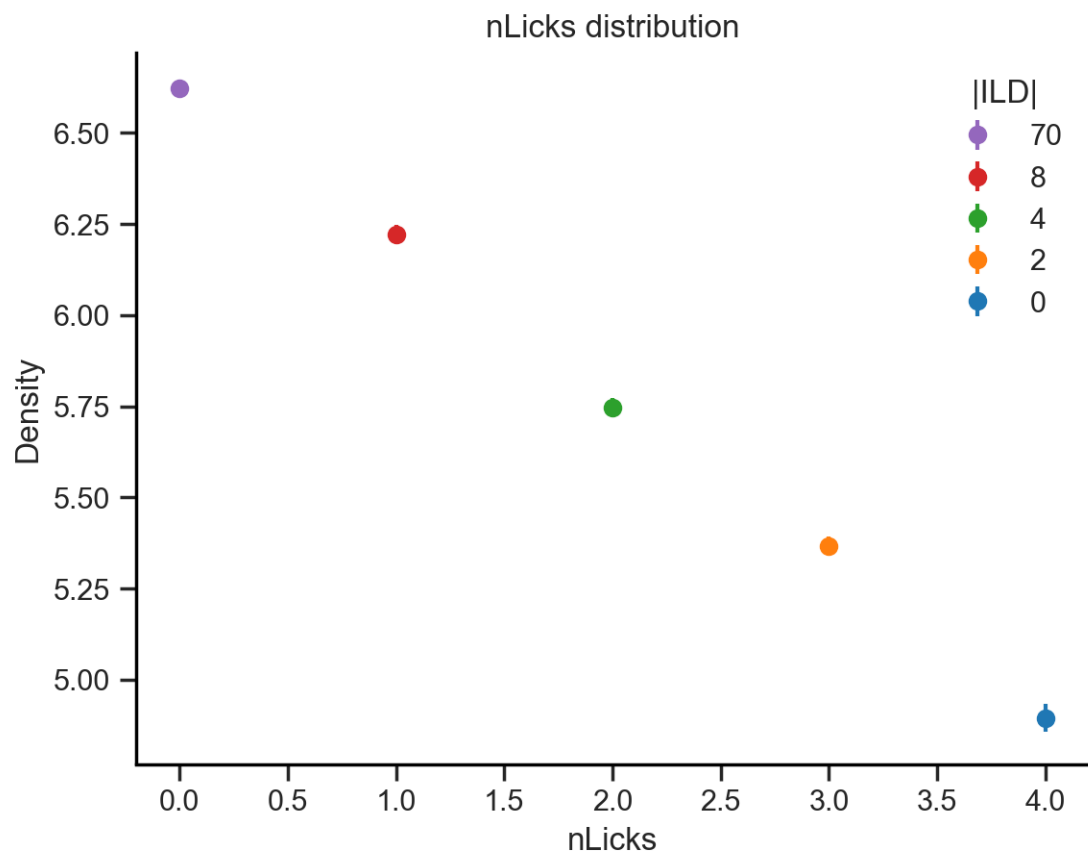
6.2 Splits

6.2.1 ILD

Corrects and errors together

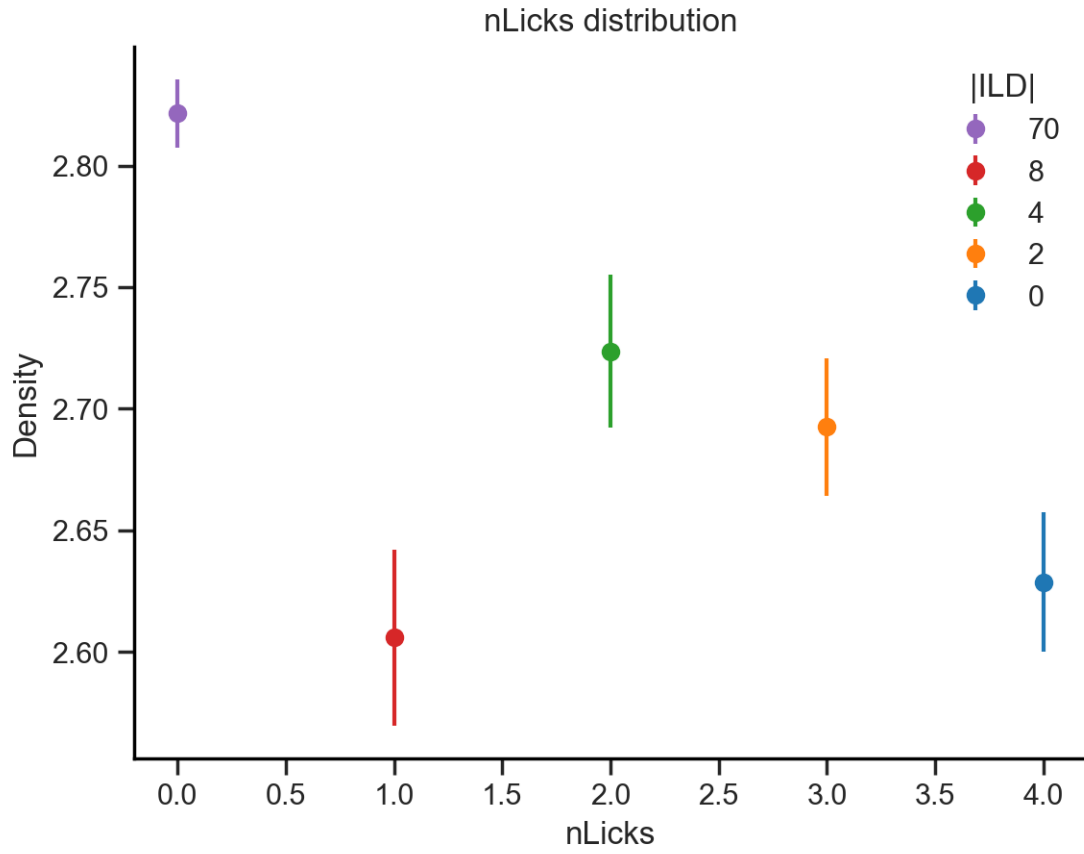
```
Mean nLicks = nan s
```

```
C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-  
packages\numpy\core\fromnumeric.py:3504: RuntimeWarning: Mean of empty slice.  
    return _methods._mean(a, axis=axis, dtype=dtype,  
C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-  
packages\numpy\core\_methods.py:129: RuntimeWarning: invalid value encountered  
in scalar divide  
    ret = ret.dtype.type(ret / rcount)
```



Errors only

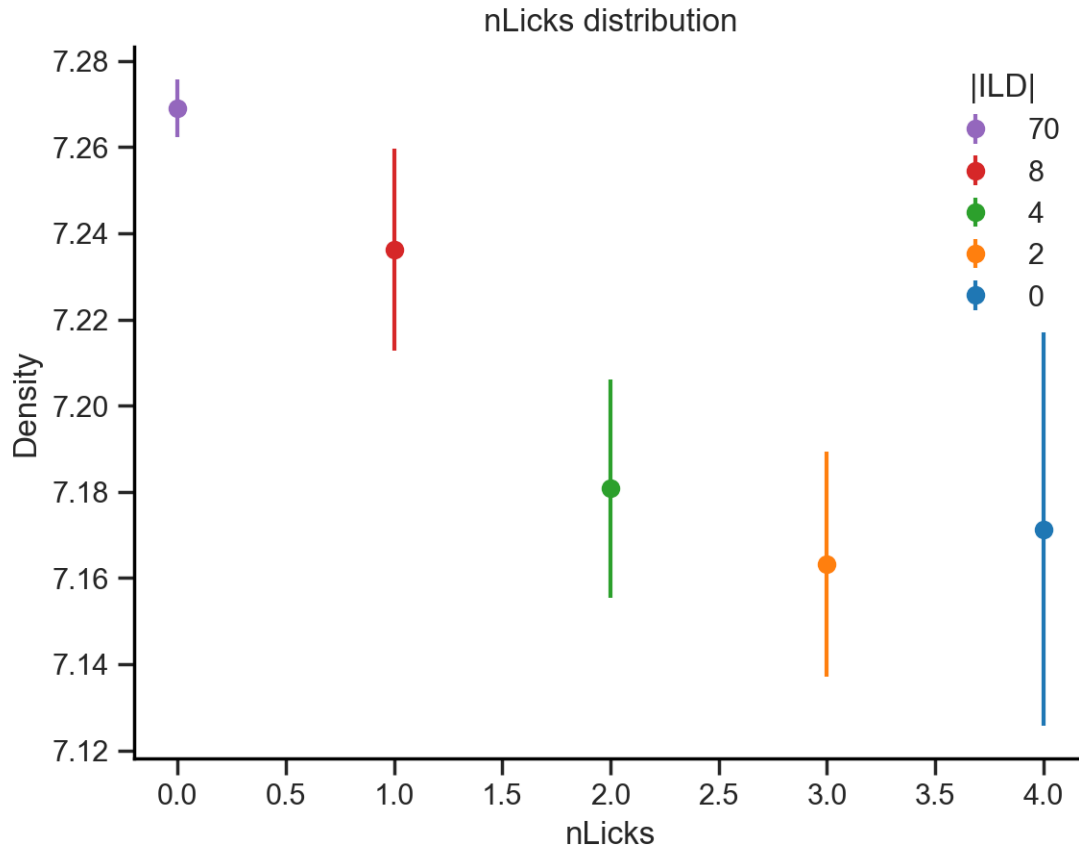
Mean nLicks = nan s



As there was no water (reward) coming out of the spouts in error trials, it makes sense that the distribution of lick rate peaks at 0. It is a needed control, but from now on the analysis will focus on **correct trials only**

Correct only

Mean nLicks = nan s



```

Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s

```

```

C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\numpy\core\_methods.py:206: RuntimeWarning: Degrees of freedom <= 0 for
slice

```

```

    ret = _var(a, axis=axis, dtype=dtype, out=out, ddof=ddof,
C:\Users\Usuario\anaconda3\envs\PycharmProjects\lib\site-
packages\numpy\core\_methods.py:198: RuntimeWarning: invalid value encountered
in scalar divide

```

```

    ret = ret.dtype.type(ret / rcount)

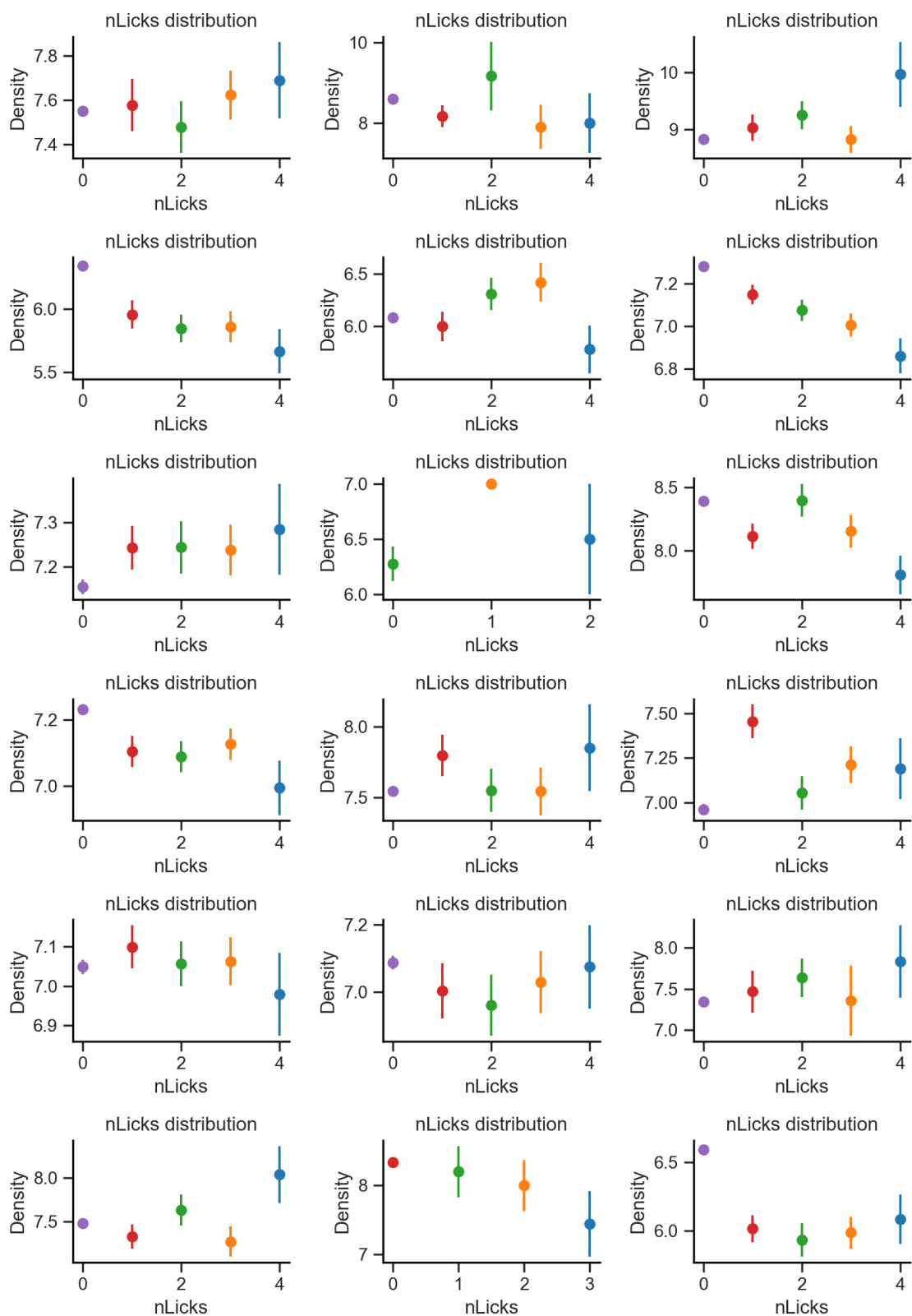
```

```

Mean nLicks = nan s
Mean nLicks = nan s

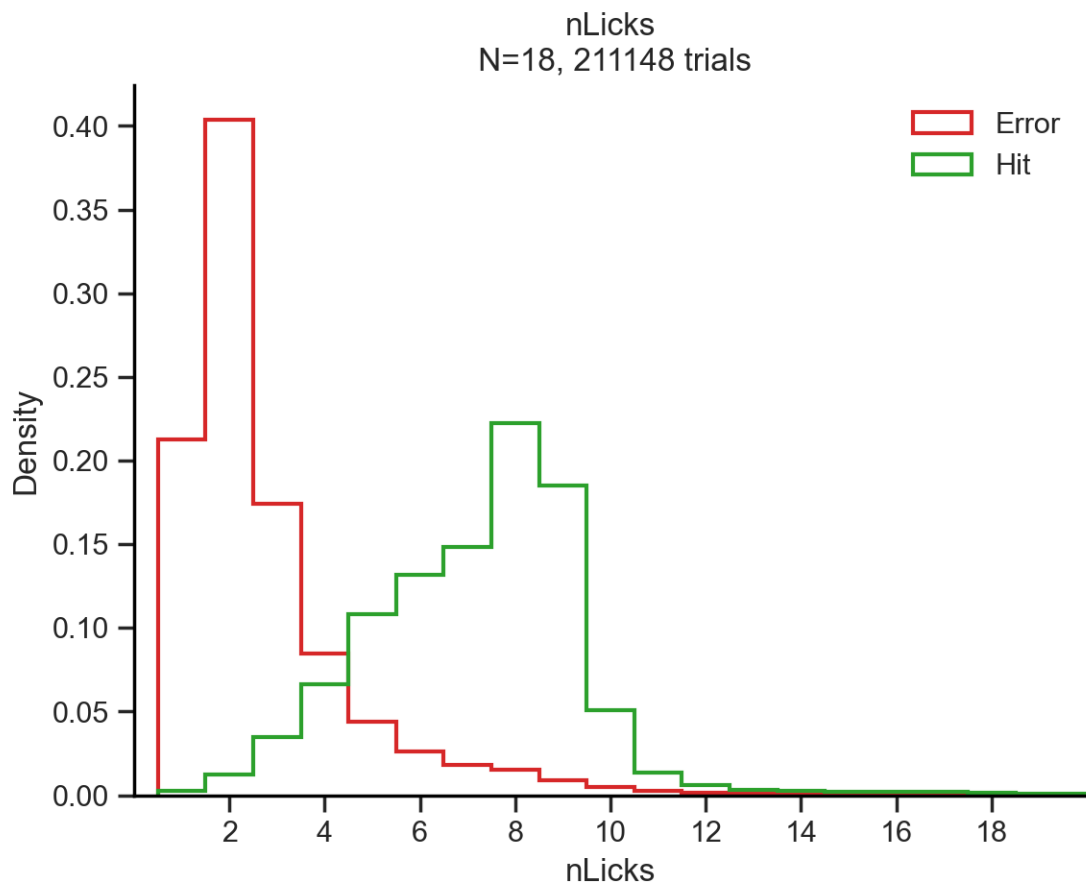
```

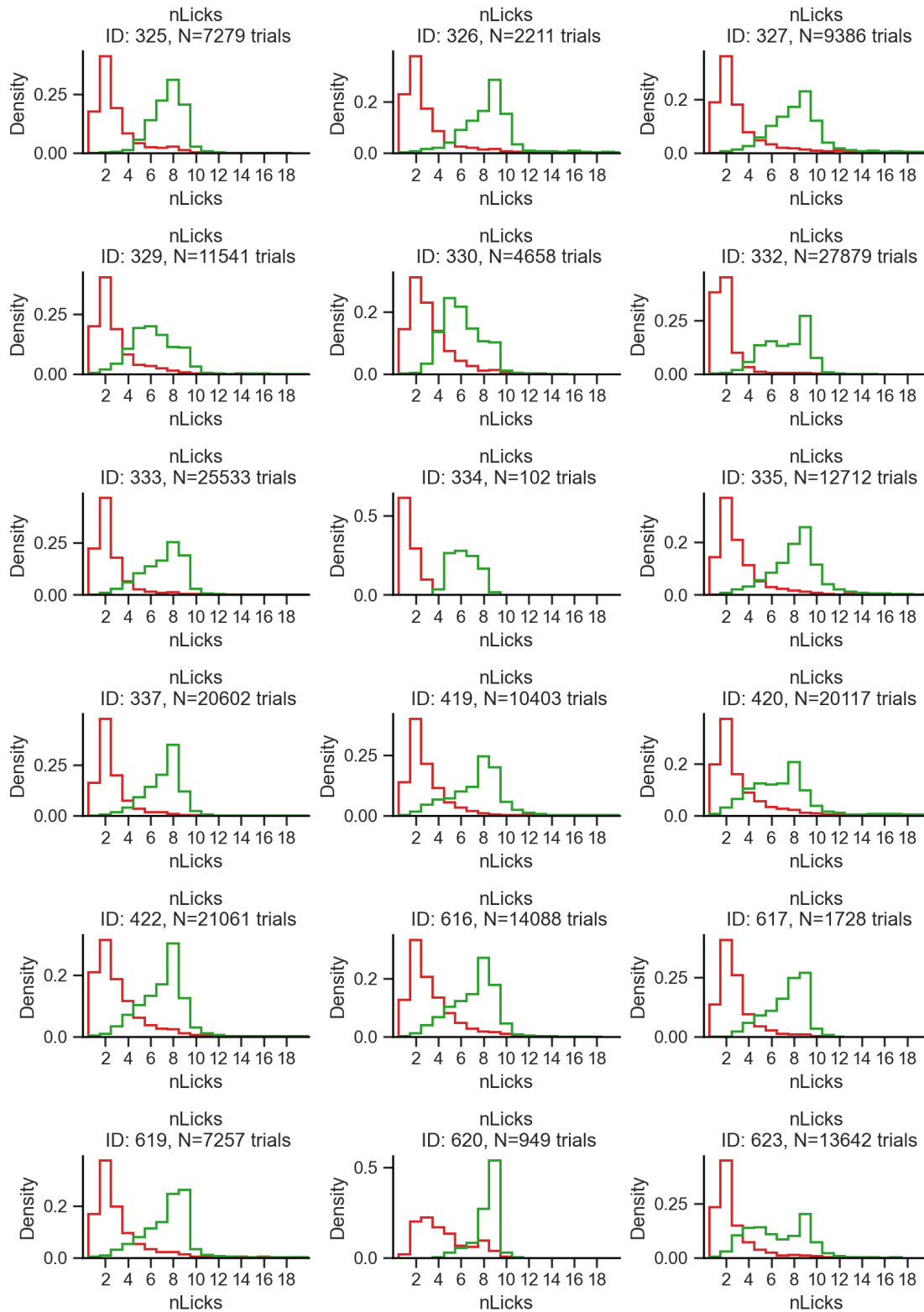
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s
Mean nLicks = nan s



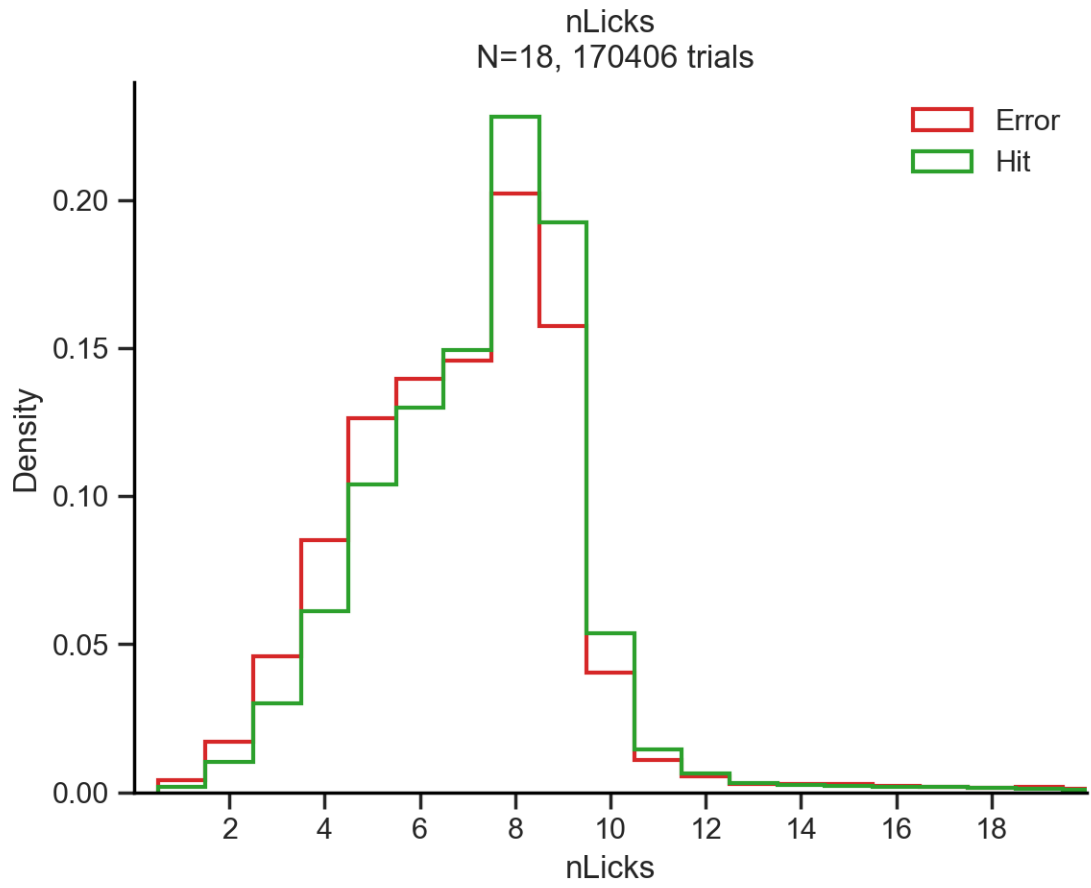
There is a modulation in the lick rate by ILD in correct trials. The higher the ILD the higher the lick rate.

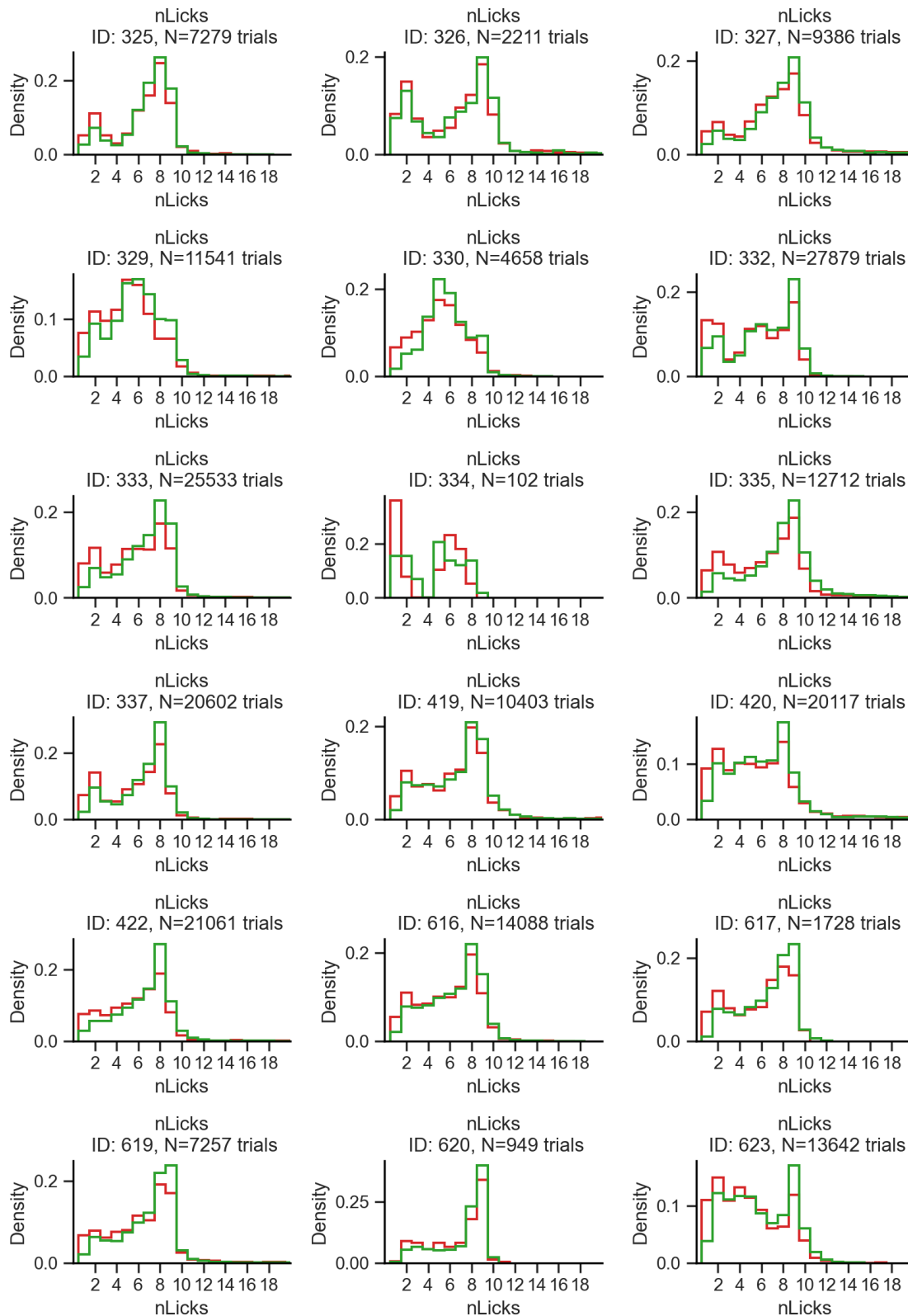
6.2.2 Outcome



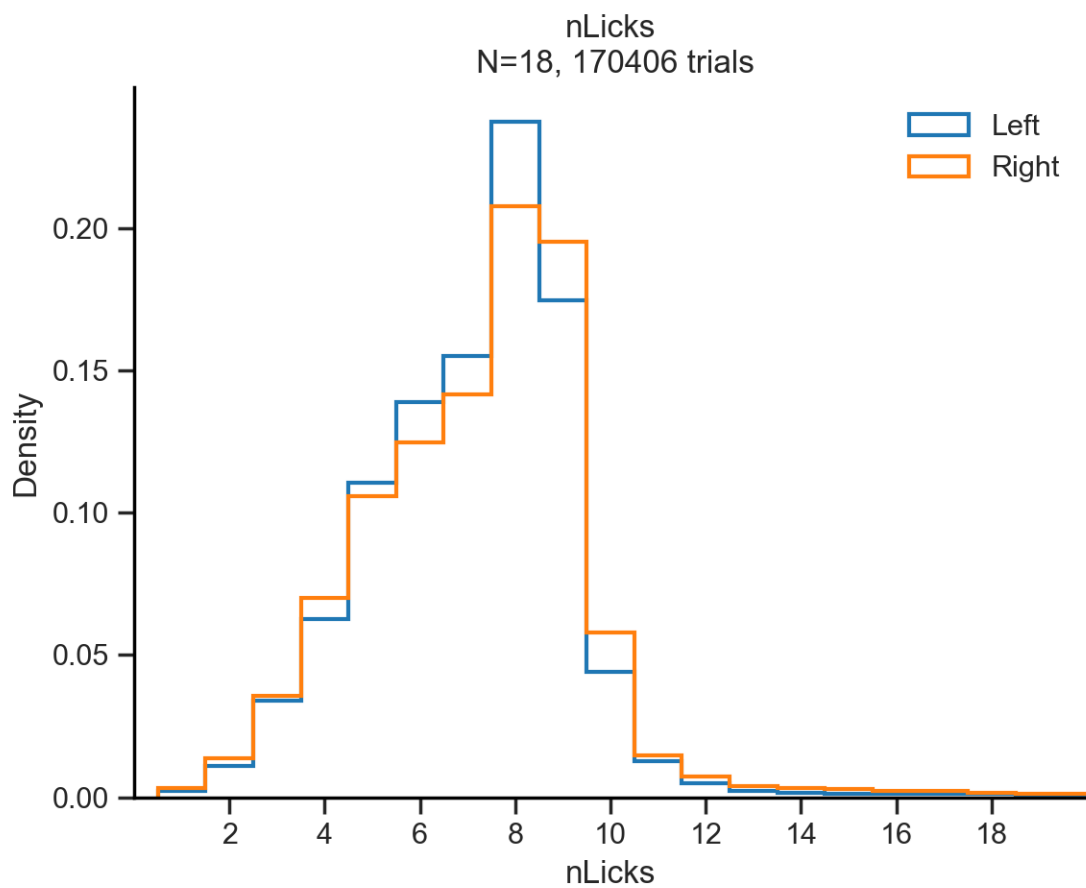


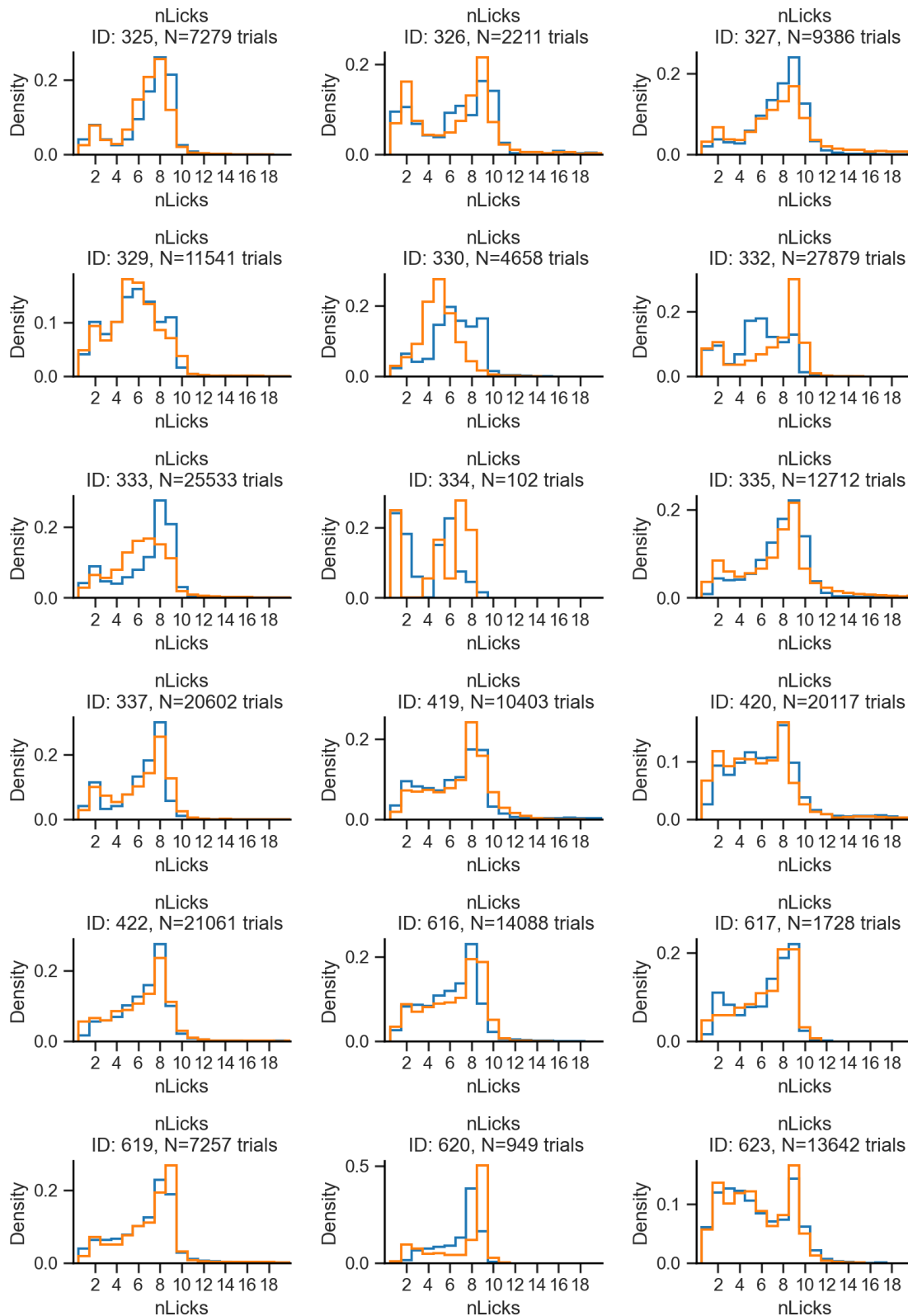
6.2.3 Previous outcome



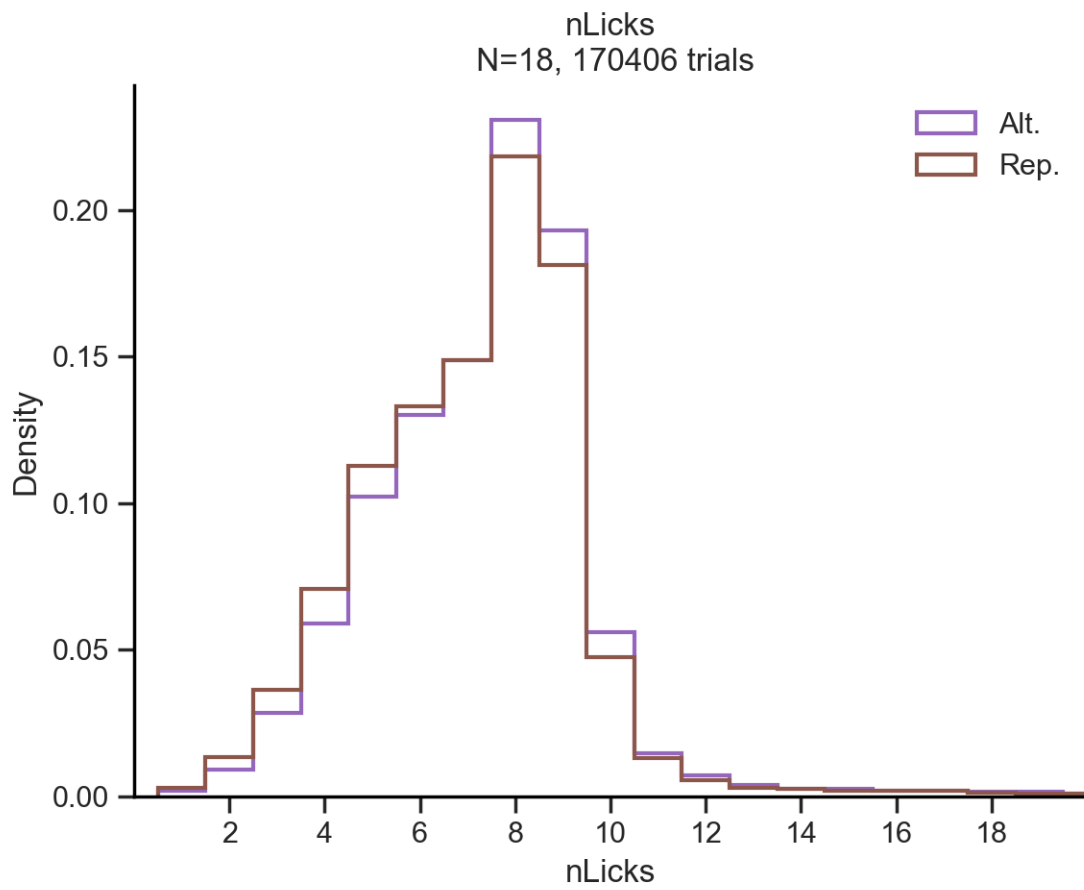


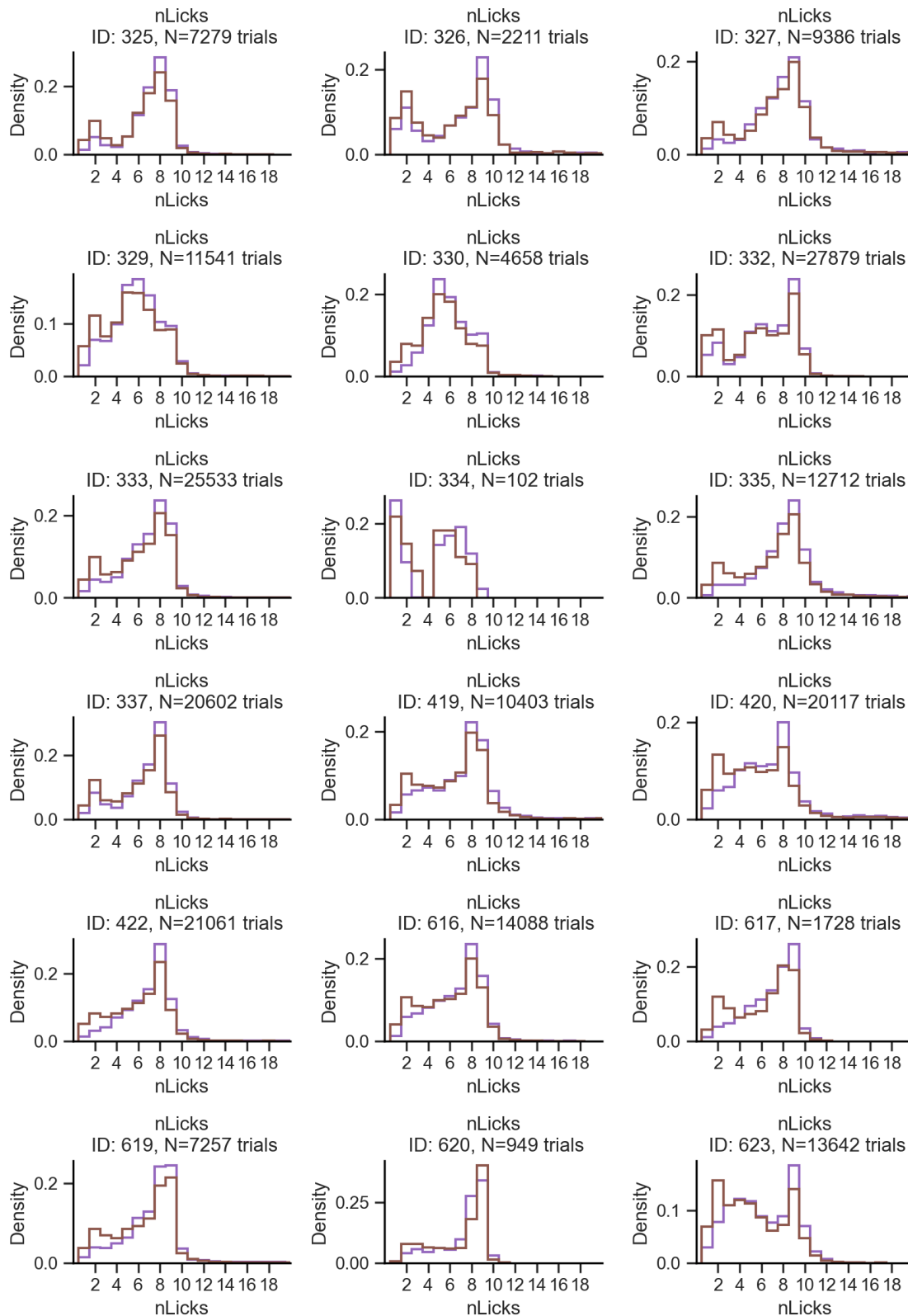
6.2.4 Choice



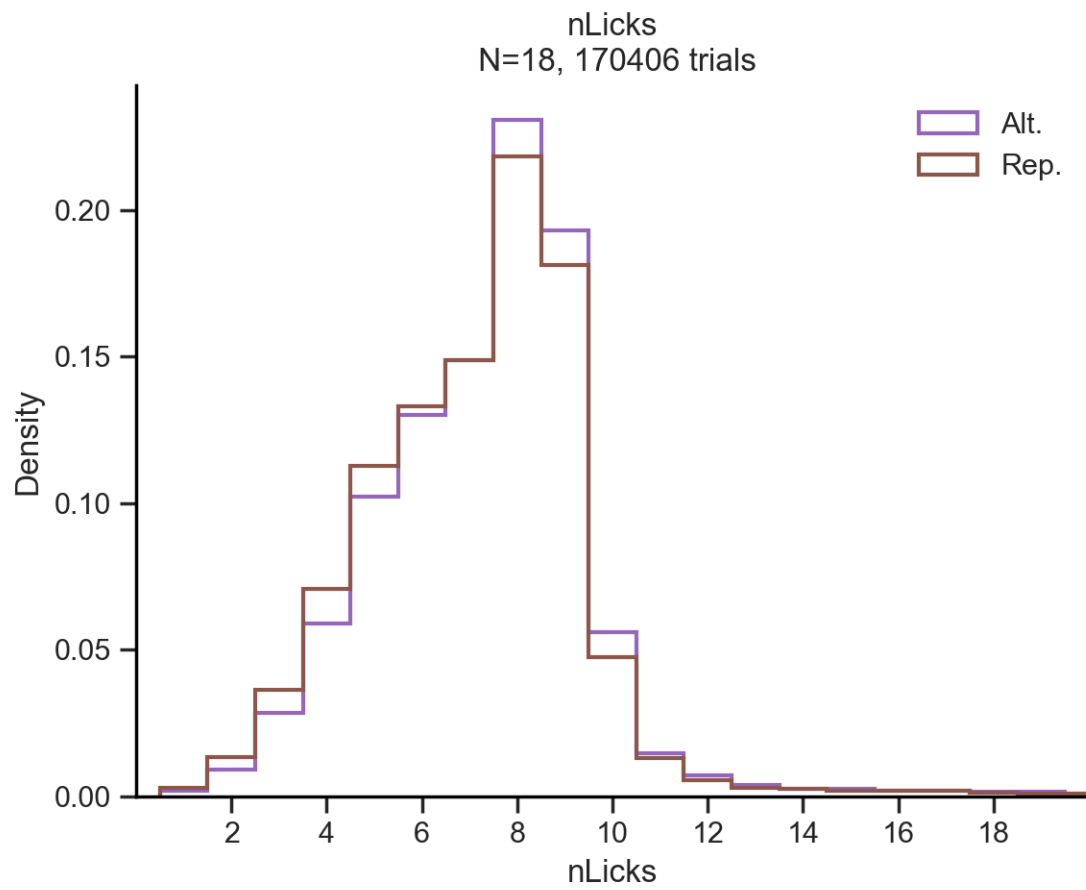


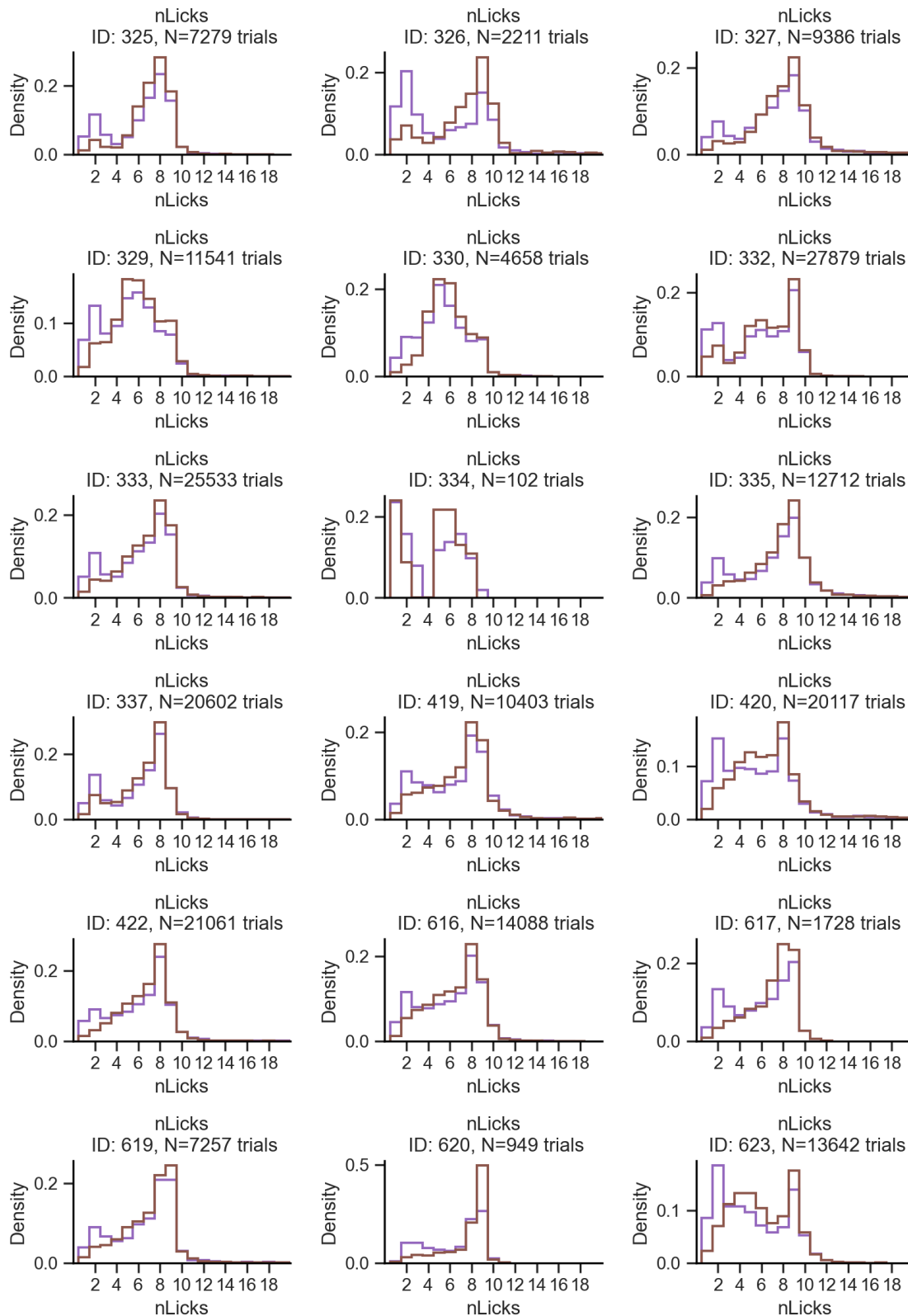
6.2.5 Repeat choice



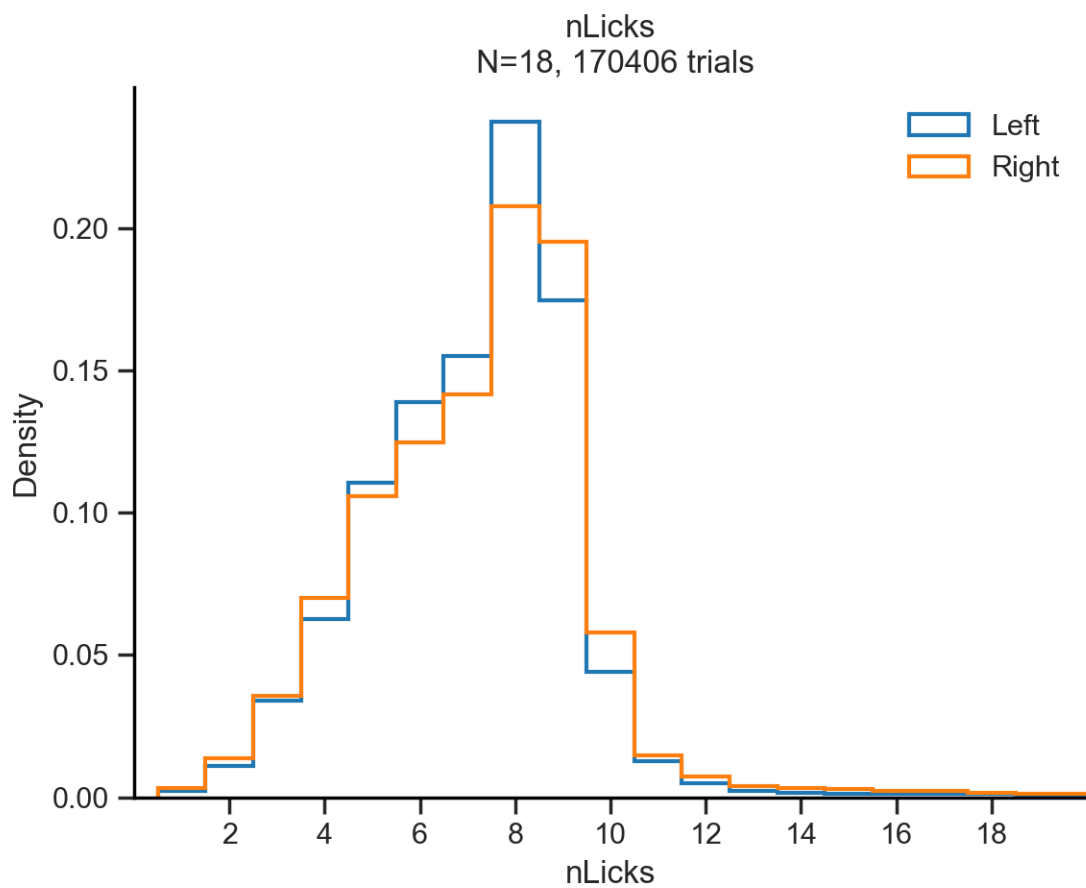


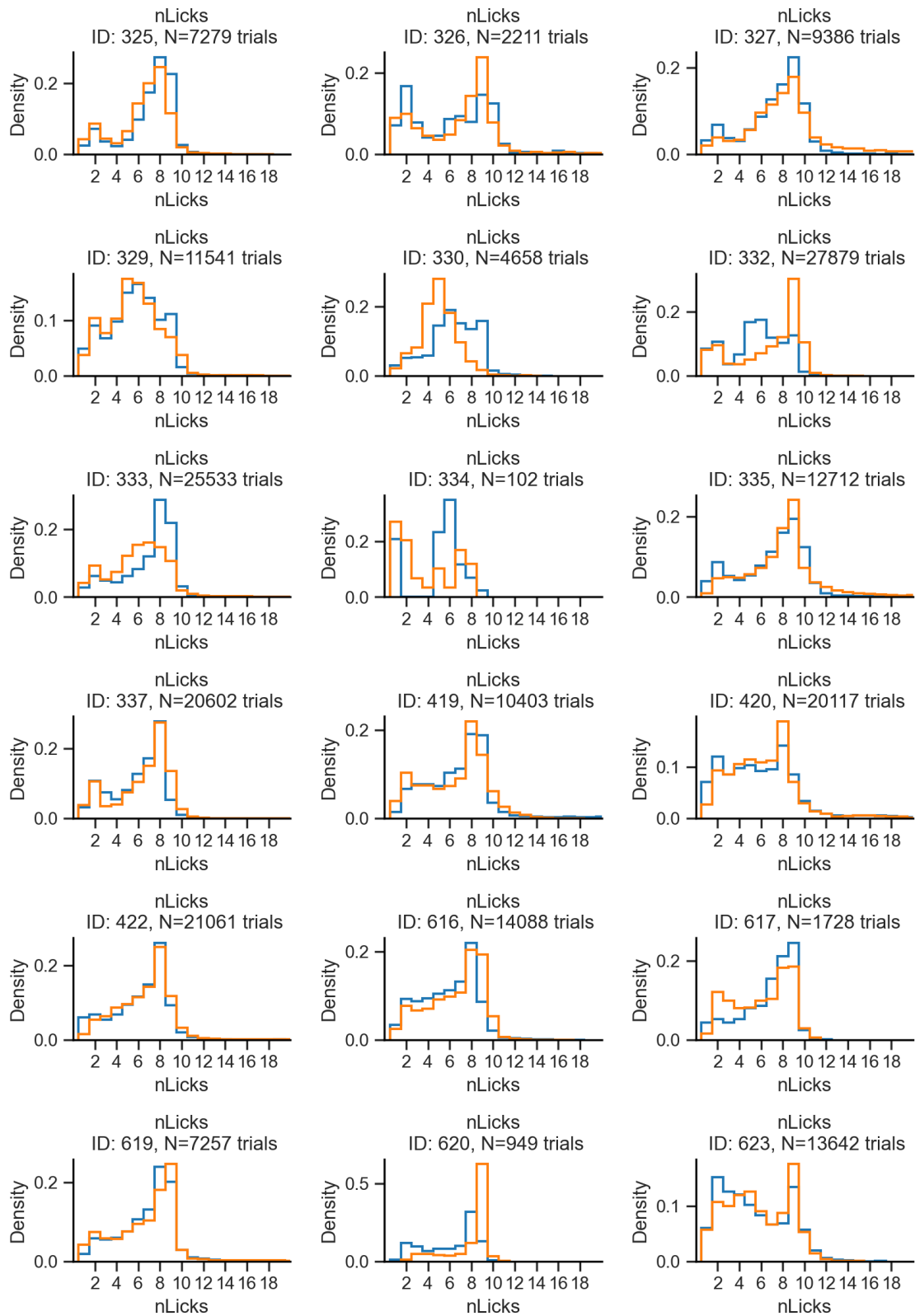
6.2.6 Repeat trial



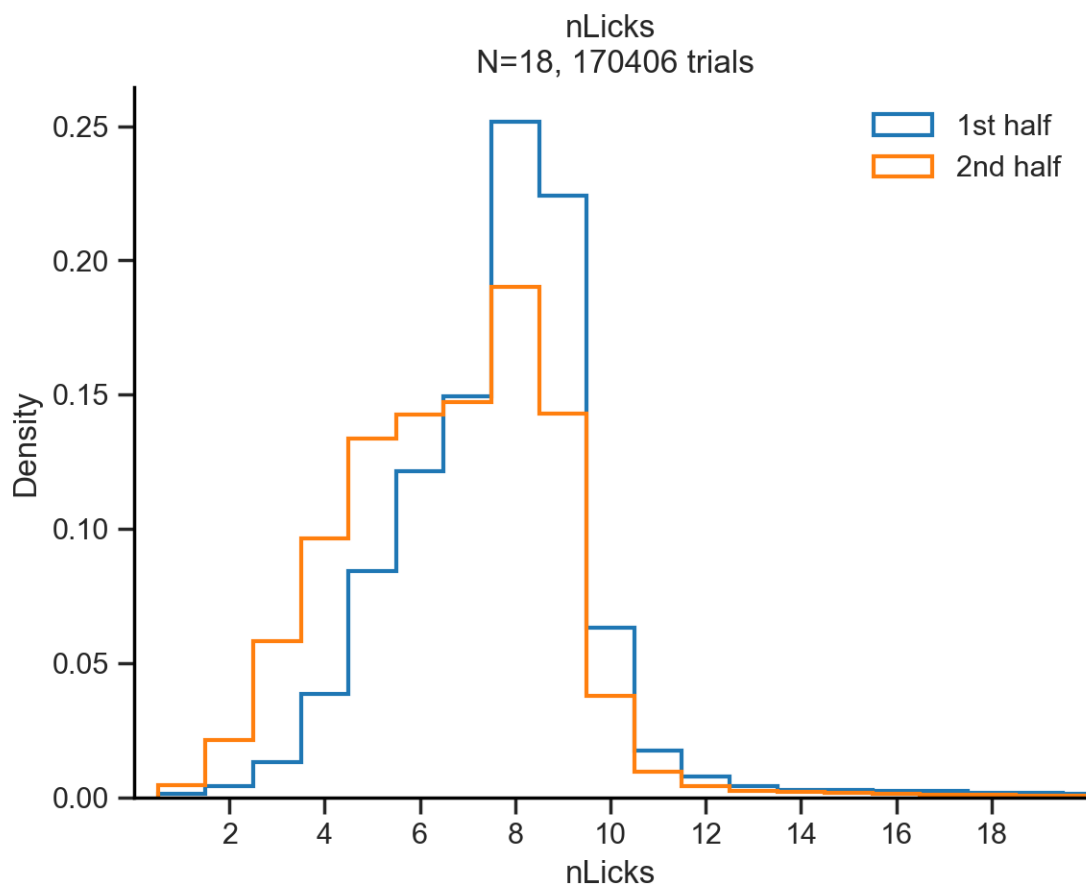


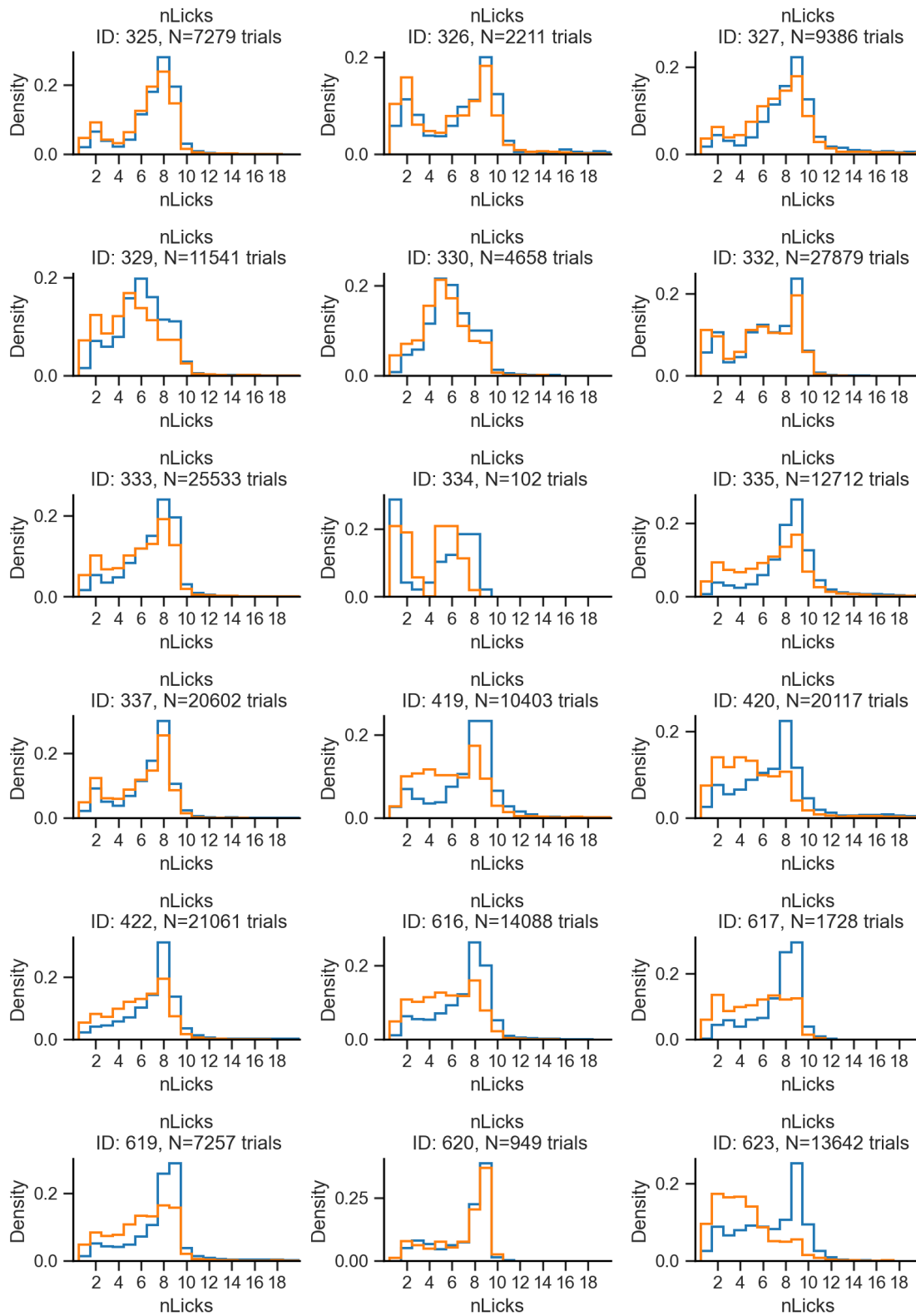
6.2.7 Stimulus





6.2.8 1st vs 2nd session half



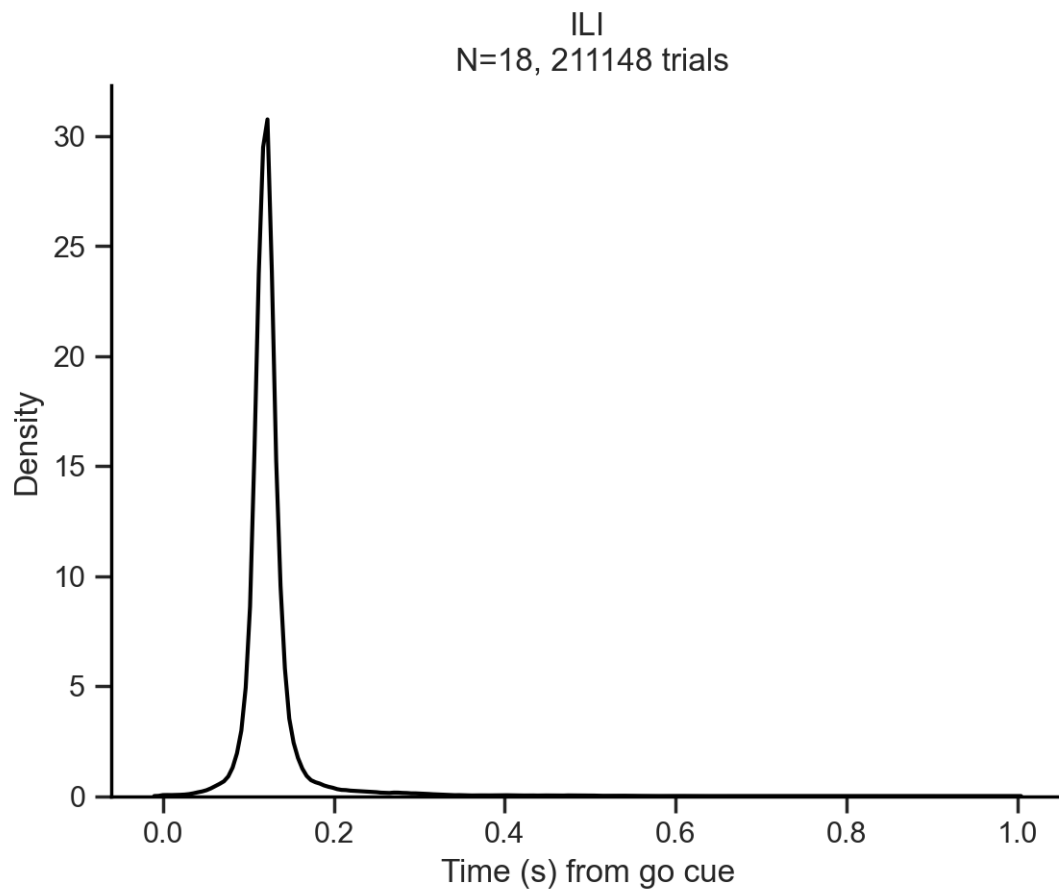


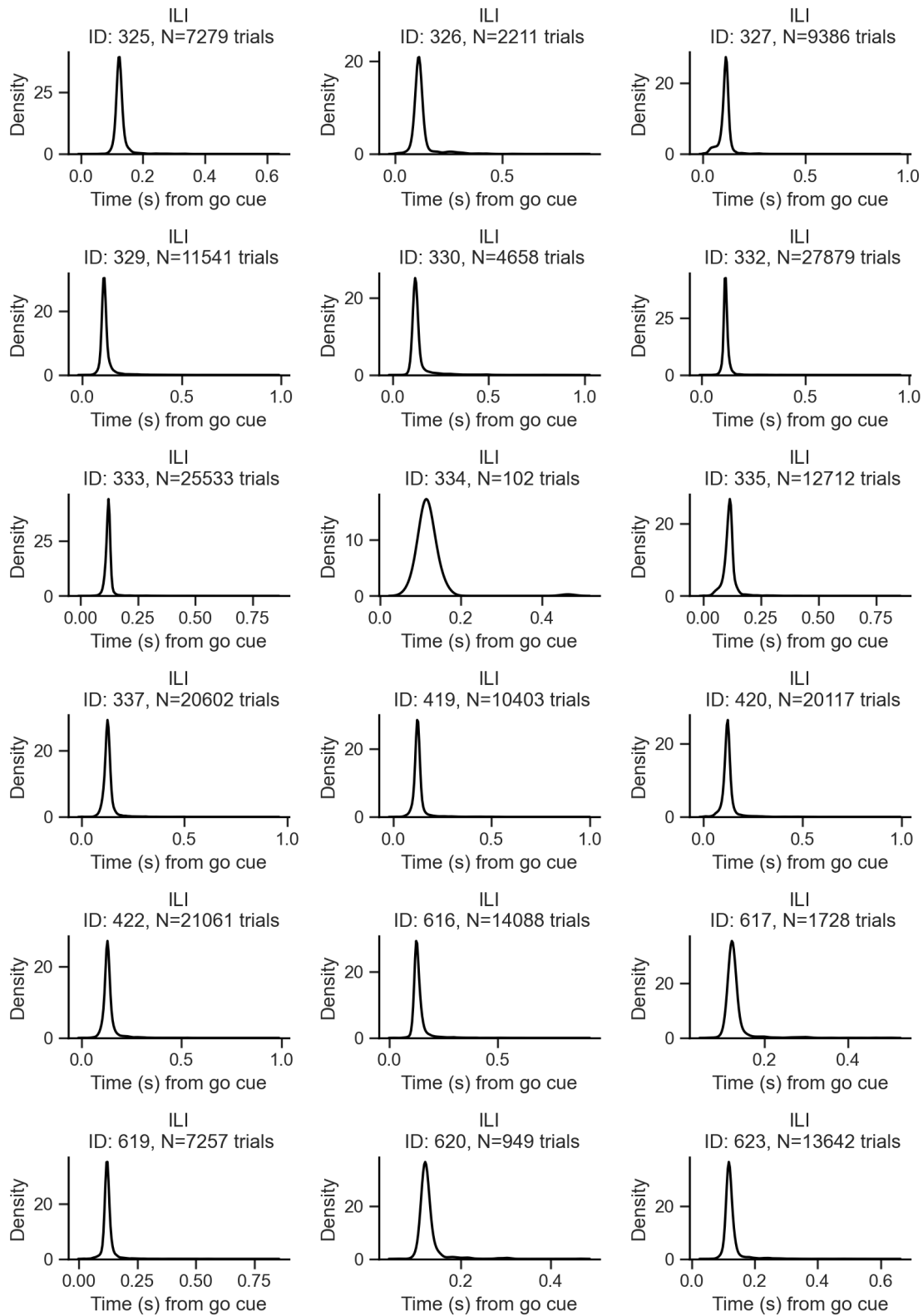
6.2.9 Hidden Markov Model (HMM)

7 Inter Lick Interval (ILI)

$$\text{Lick rate (Hz)} \approx \frac{1}{\text{Mean ILI (s)}}$$

7.0.1 All trials

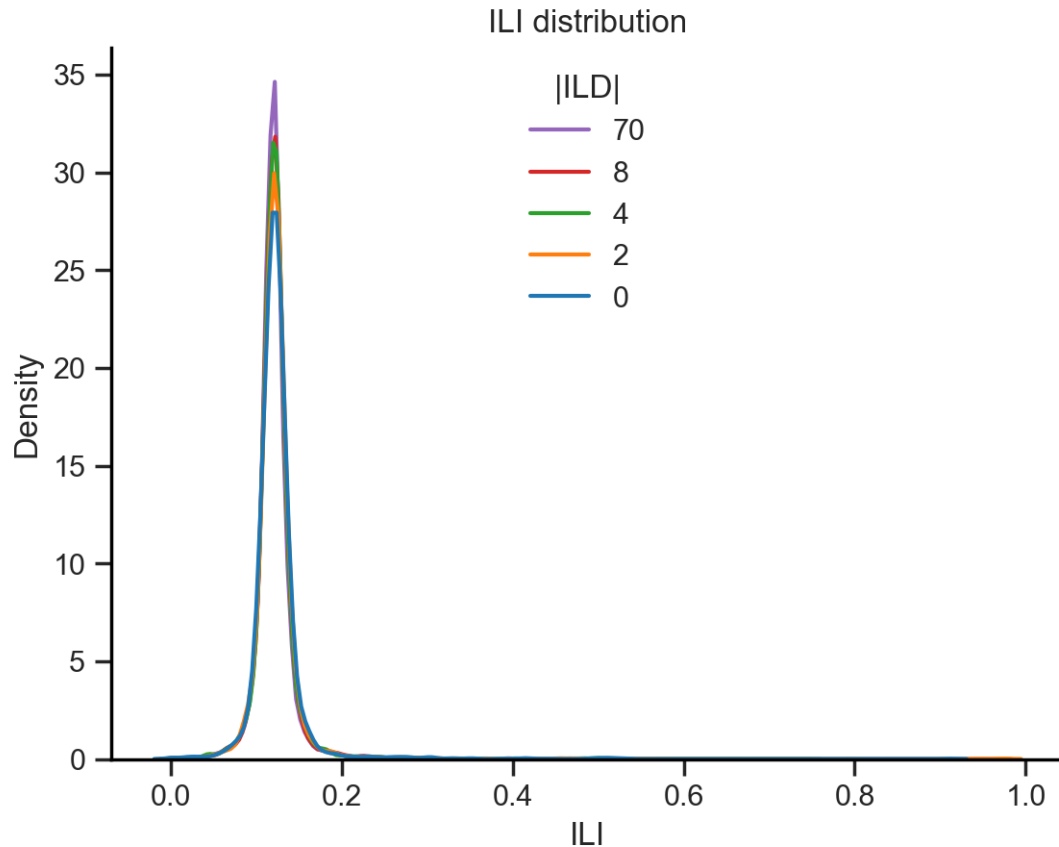




7.0.2 Splits

ILD

Mean ILI = 0.121 s



Mean ILI = 0.123 s

Mean ILI = 0.110 s

Mean ILI = 0.113 s

Mean ILI = 0.110 s

Mean ILI = 0.119 s

Mean ILI = 0.115 s

Mean ILI = 0.121 s

Mean ILI = 0.117 s

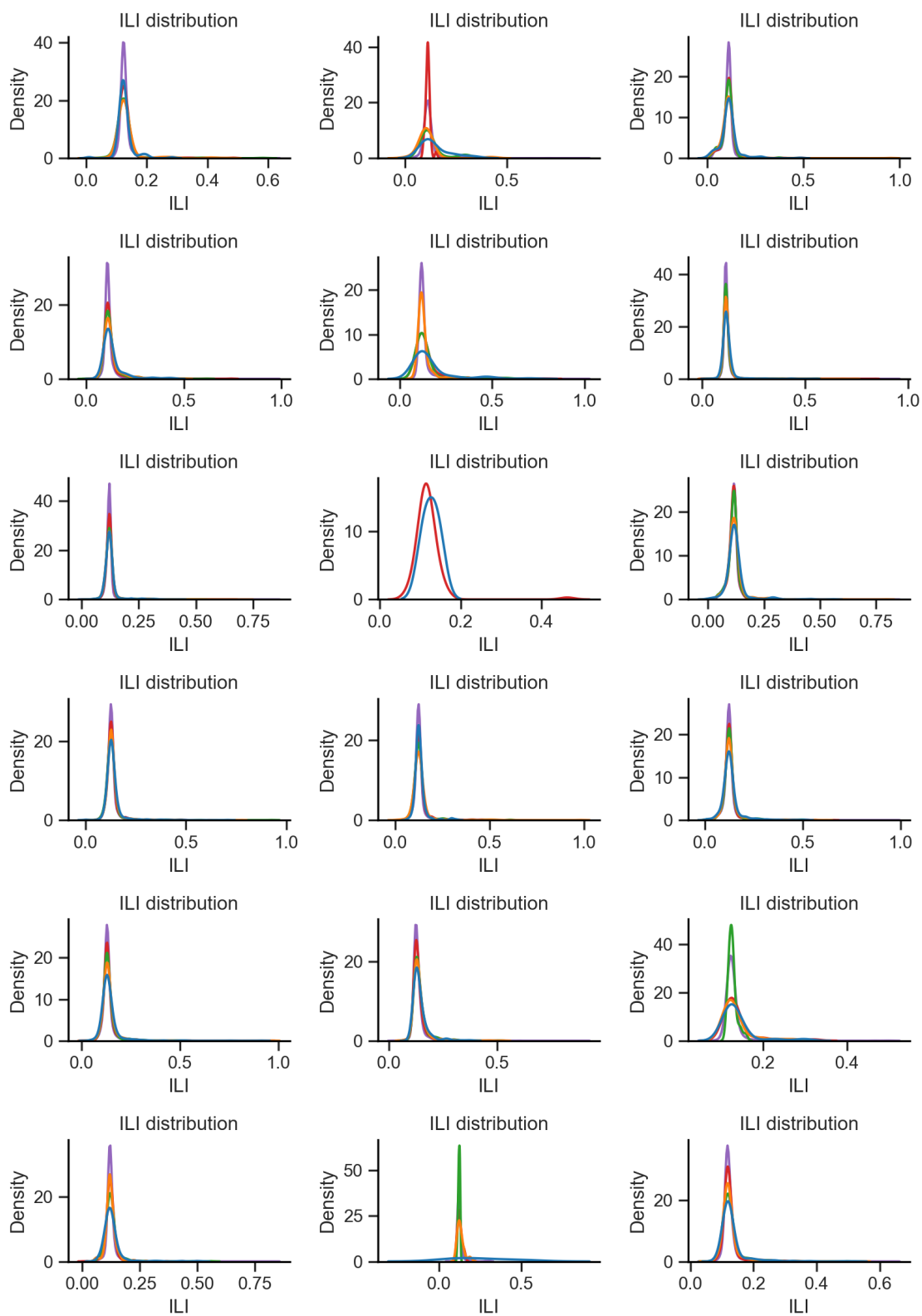
Mean ILI = 0.117 s

C:\Users\Usuario\PycharmProjects\licks\licks.py:395: UserWarning: Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this warning.

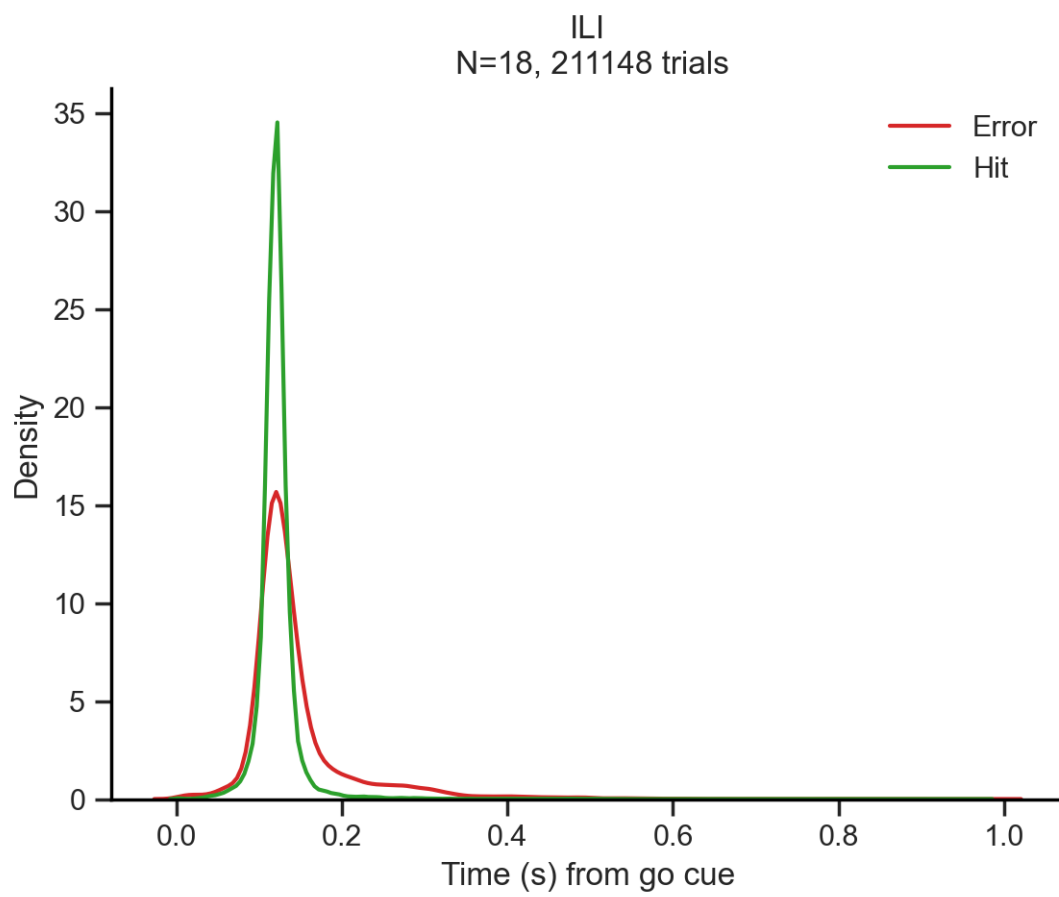
```
line = plt.gca().lines[-1]
```

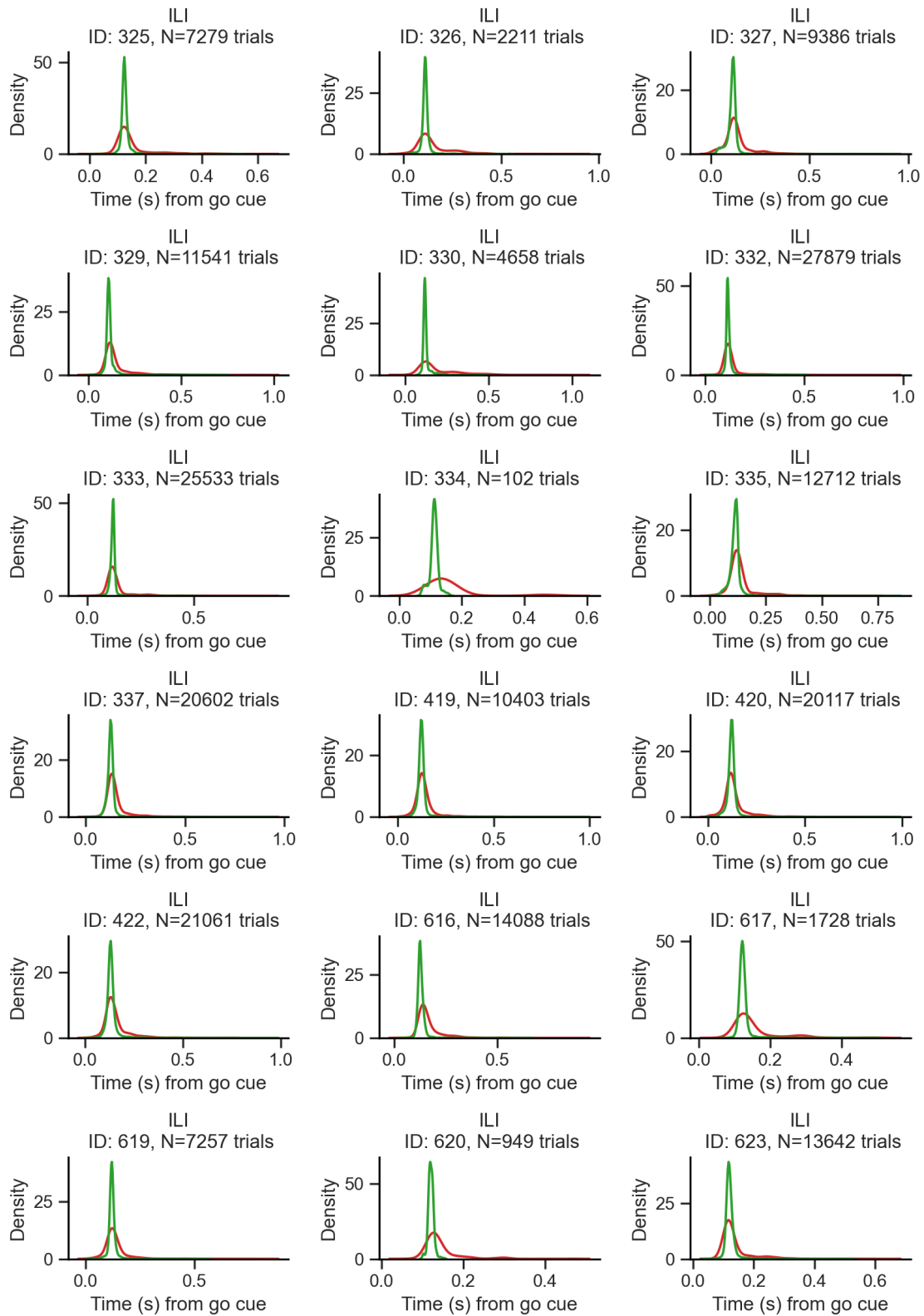
C:\Users\Usuario\PycharmProjects\licks\licks.py:395: UserWarning: Dataset has 0 variance; skipping density estimate. Pass `warn\_singular=False` to disable this


```
warning.  
    line = plt.gca().lines[-1]  
  
Mean ILI = 0.128 s  
Mean ILI = 0.124 s  
Mean ILI = 0.123 s  
Mean ILI = 0.129 s  
Mean ILI = 0.127 s  
Mean ILI = 0.123 s  
Mean ILI = 0.121 s  
Mean ILI = 0.131 s  
Mean ILI = 0.119 s
```

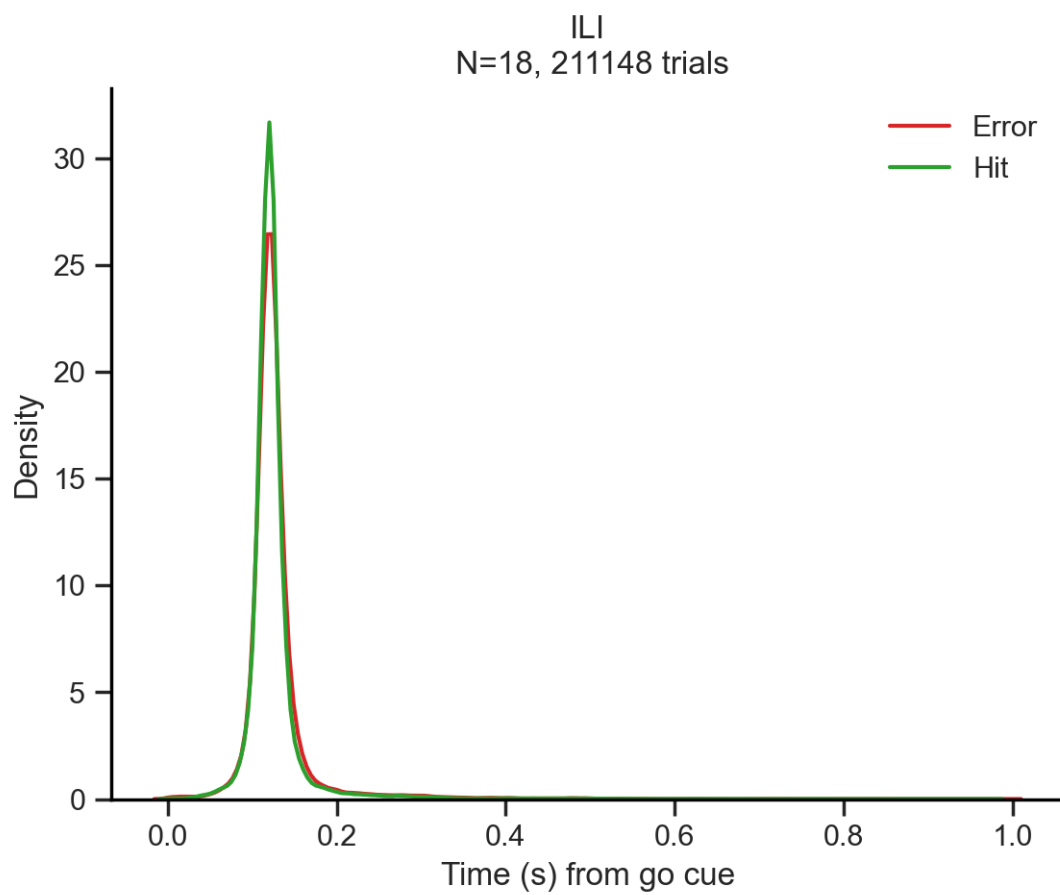


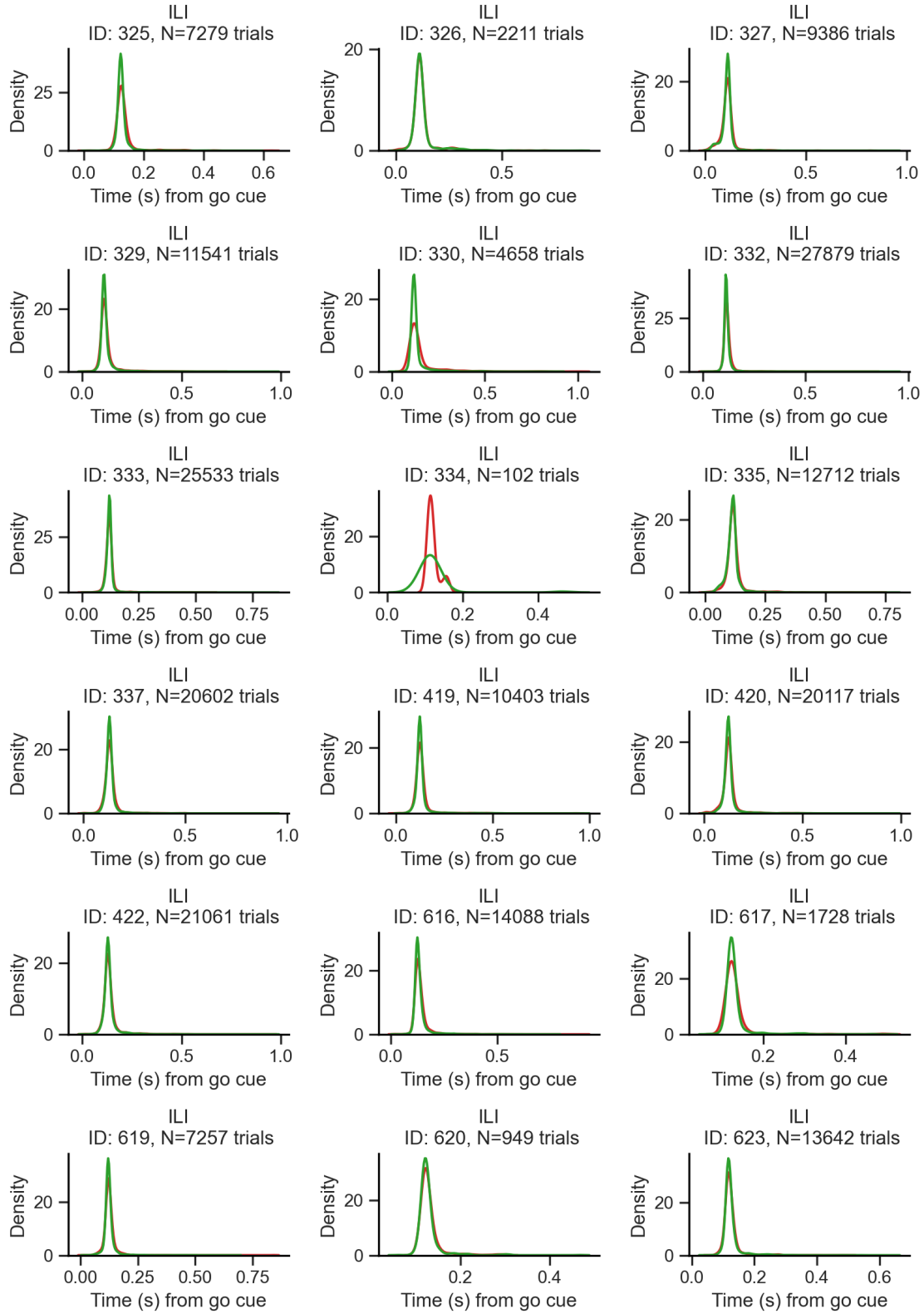
Outcome



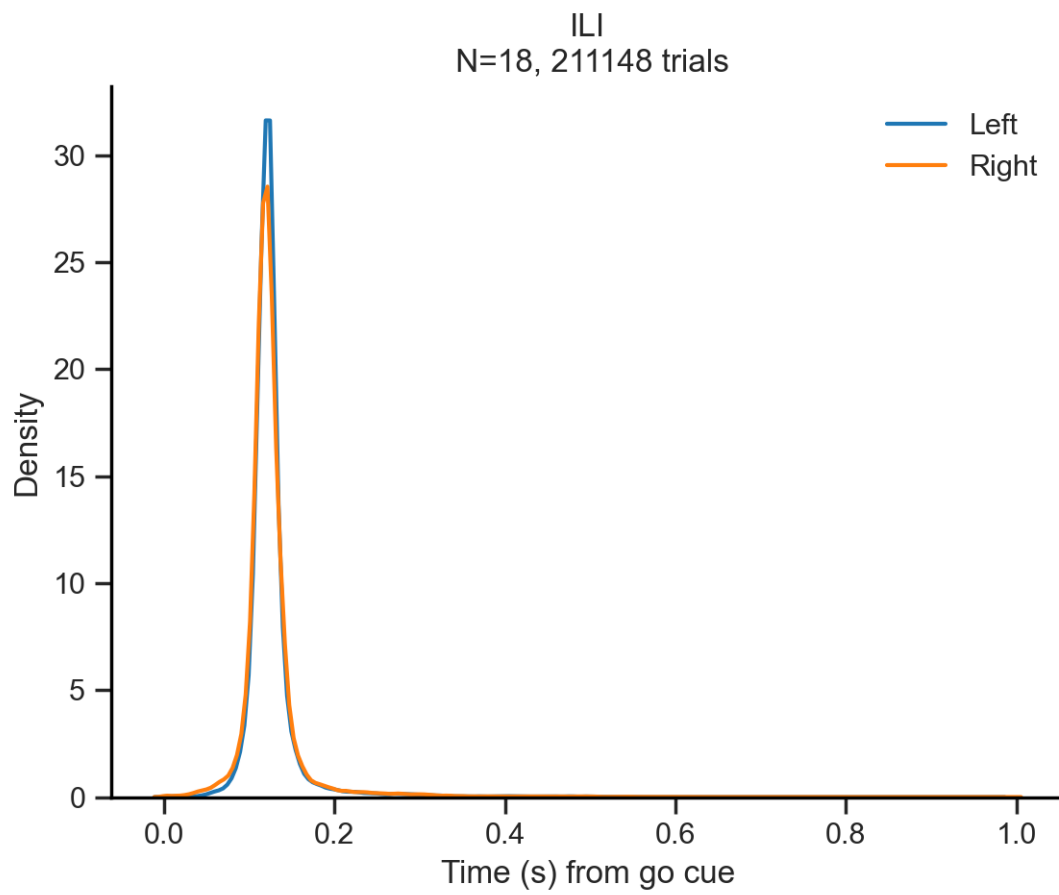


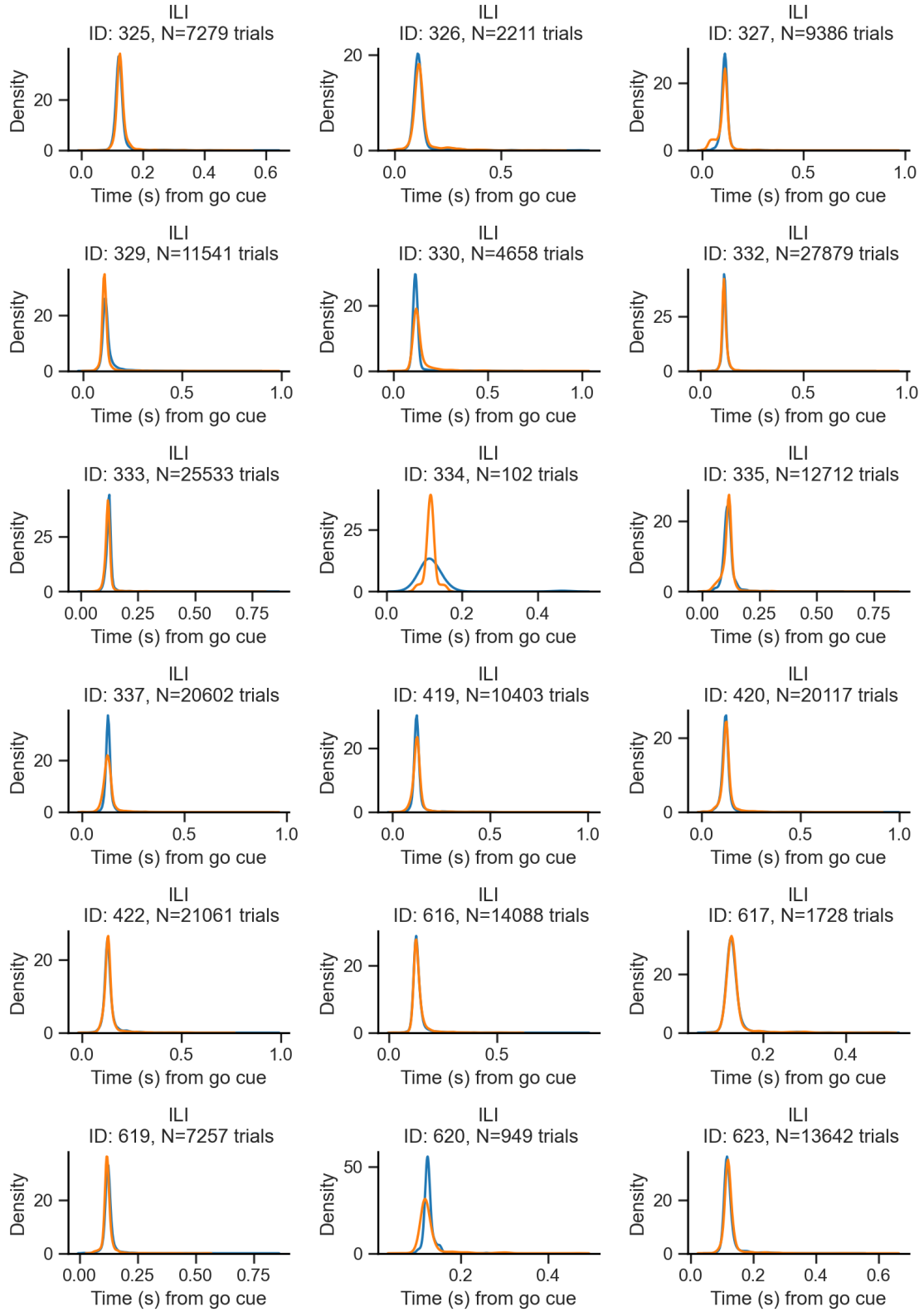
Previous Outcome



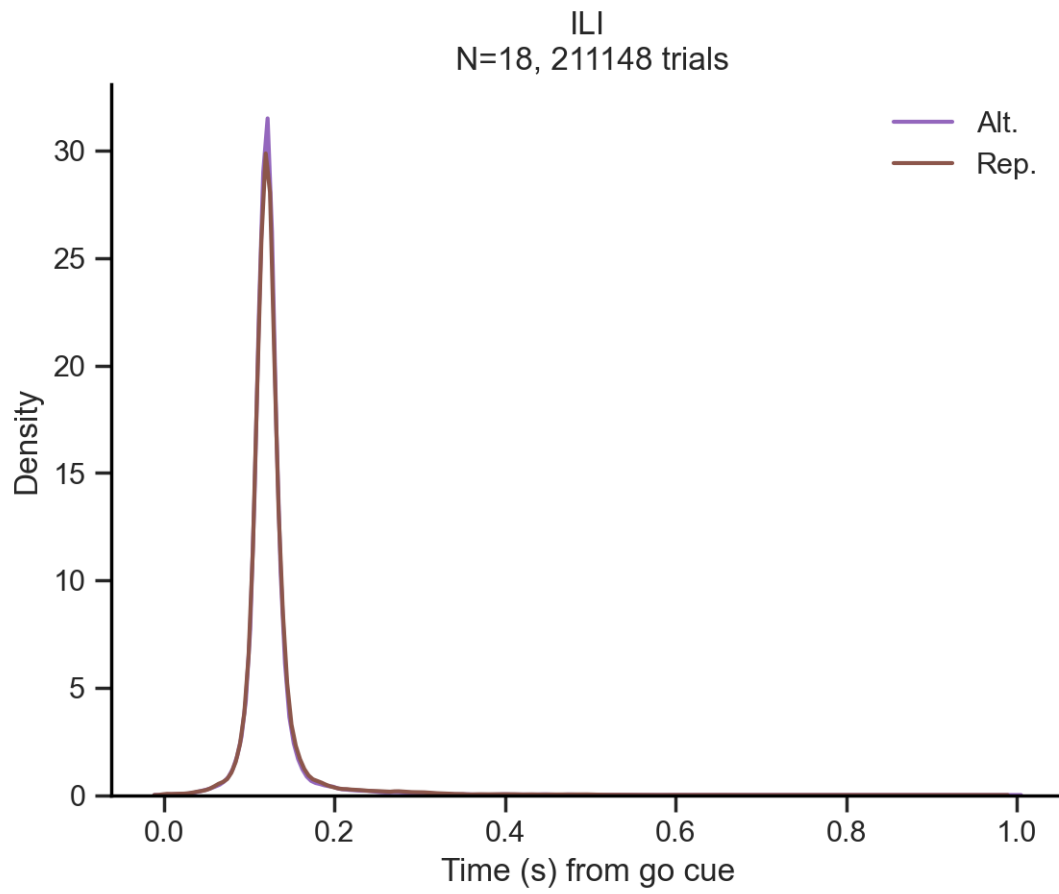


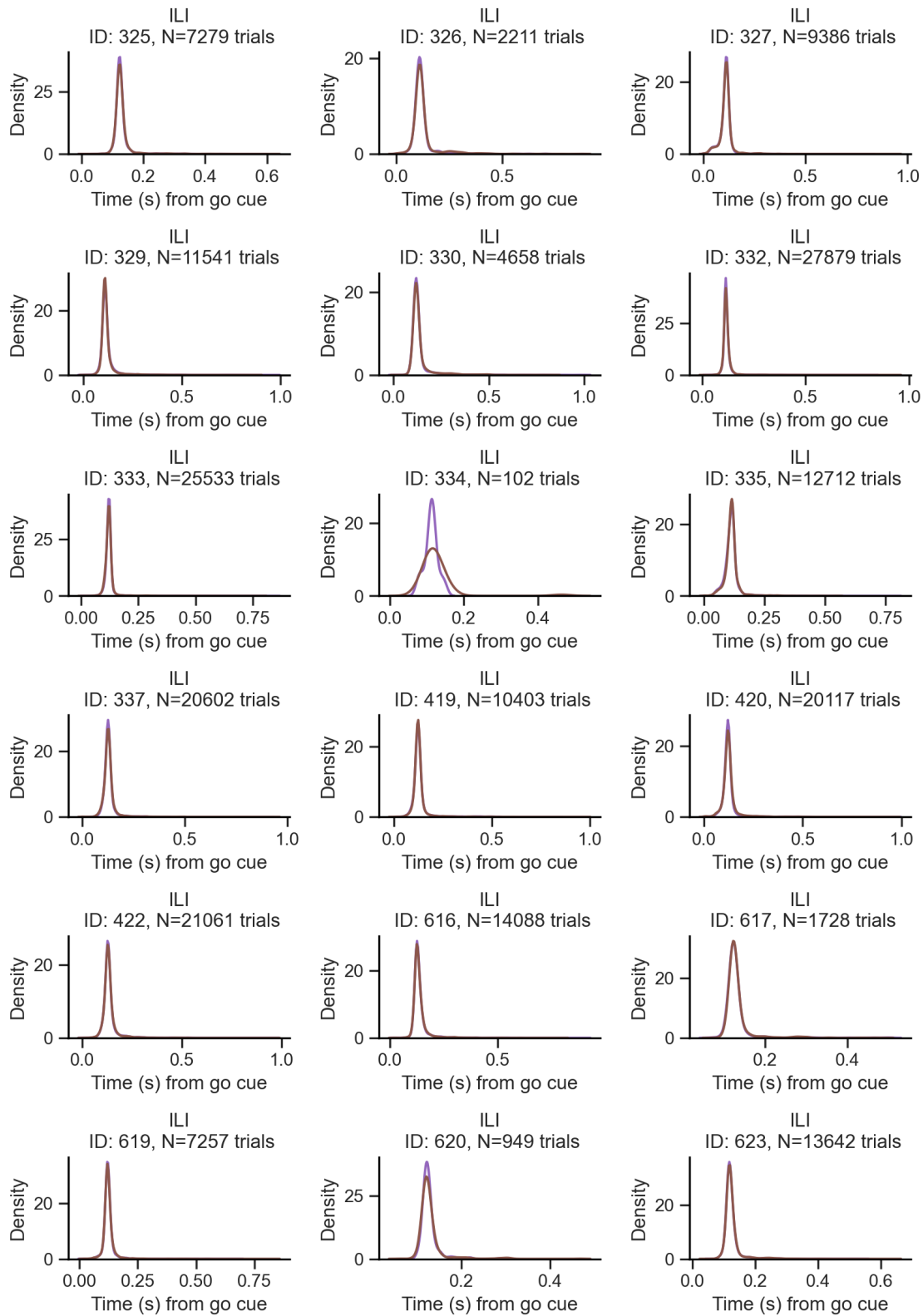
Choice



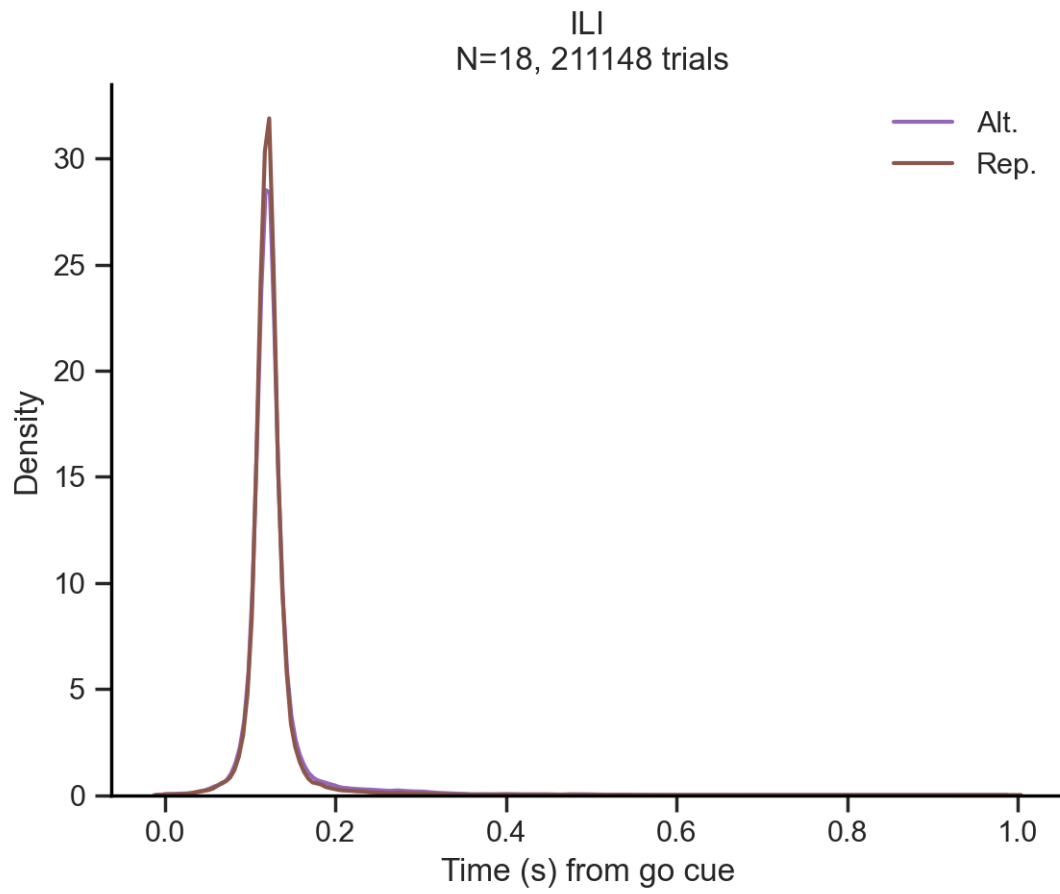


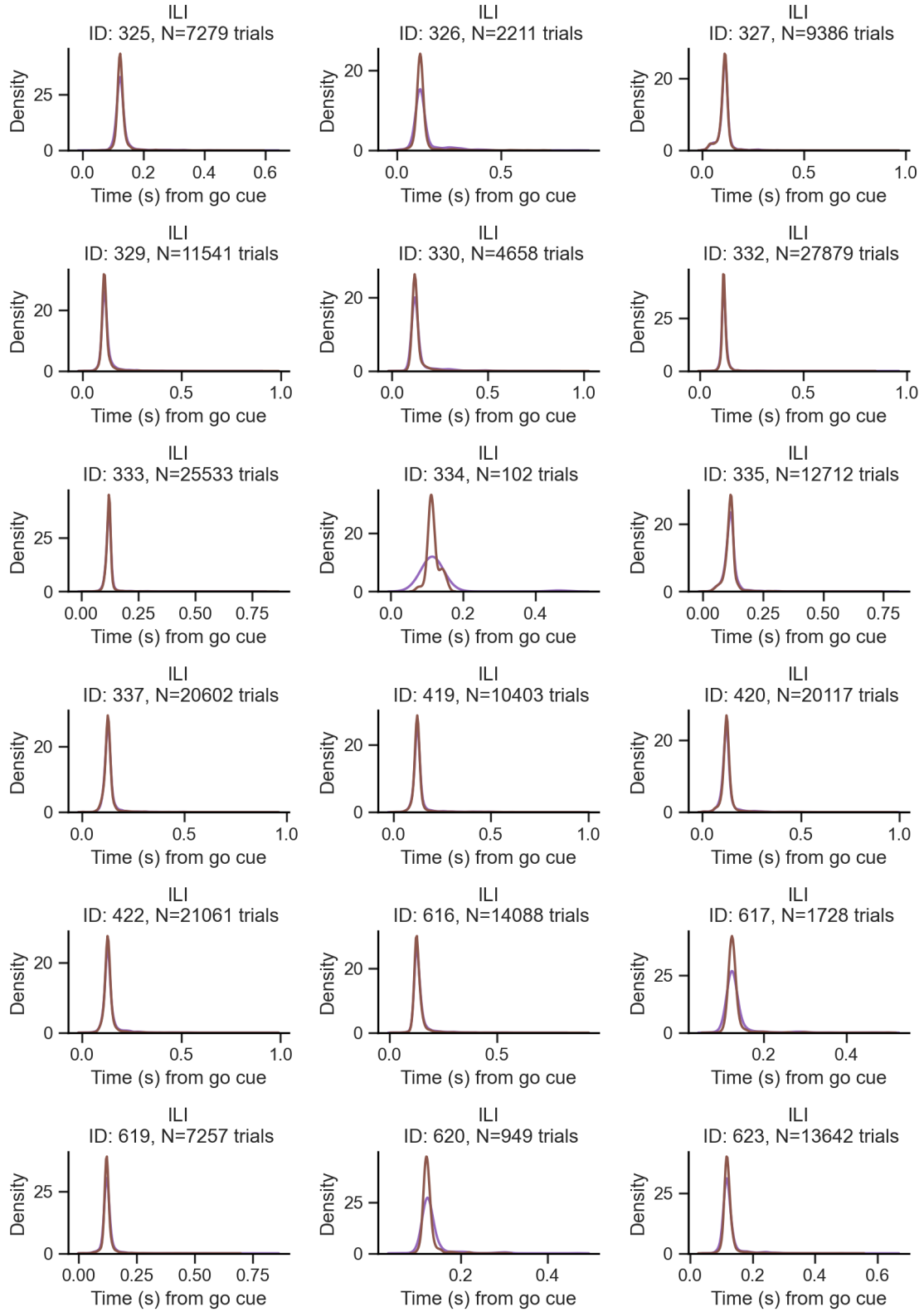
Repeat choice



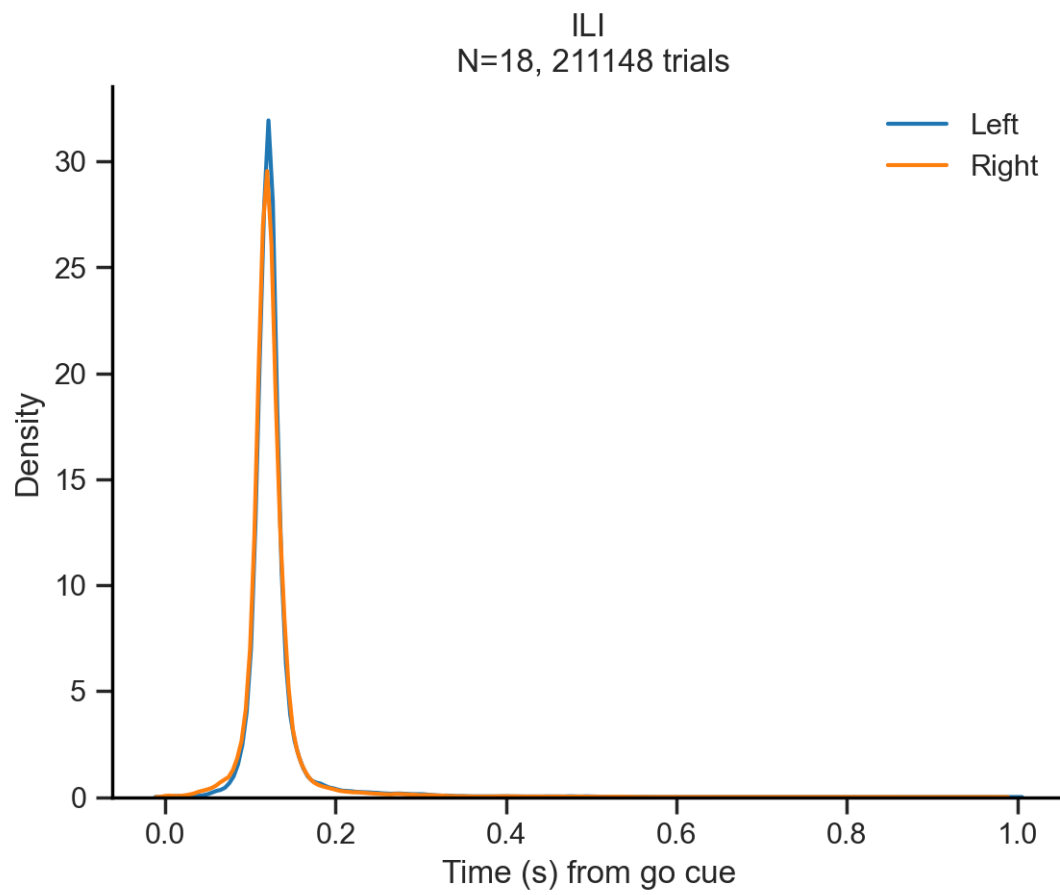


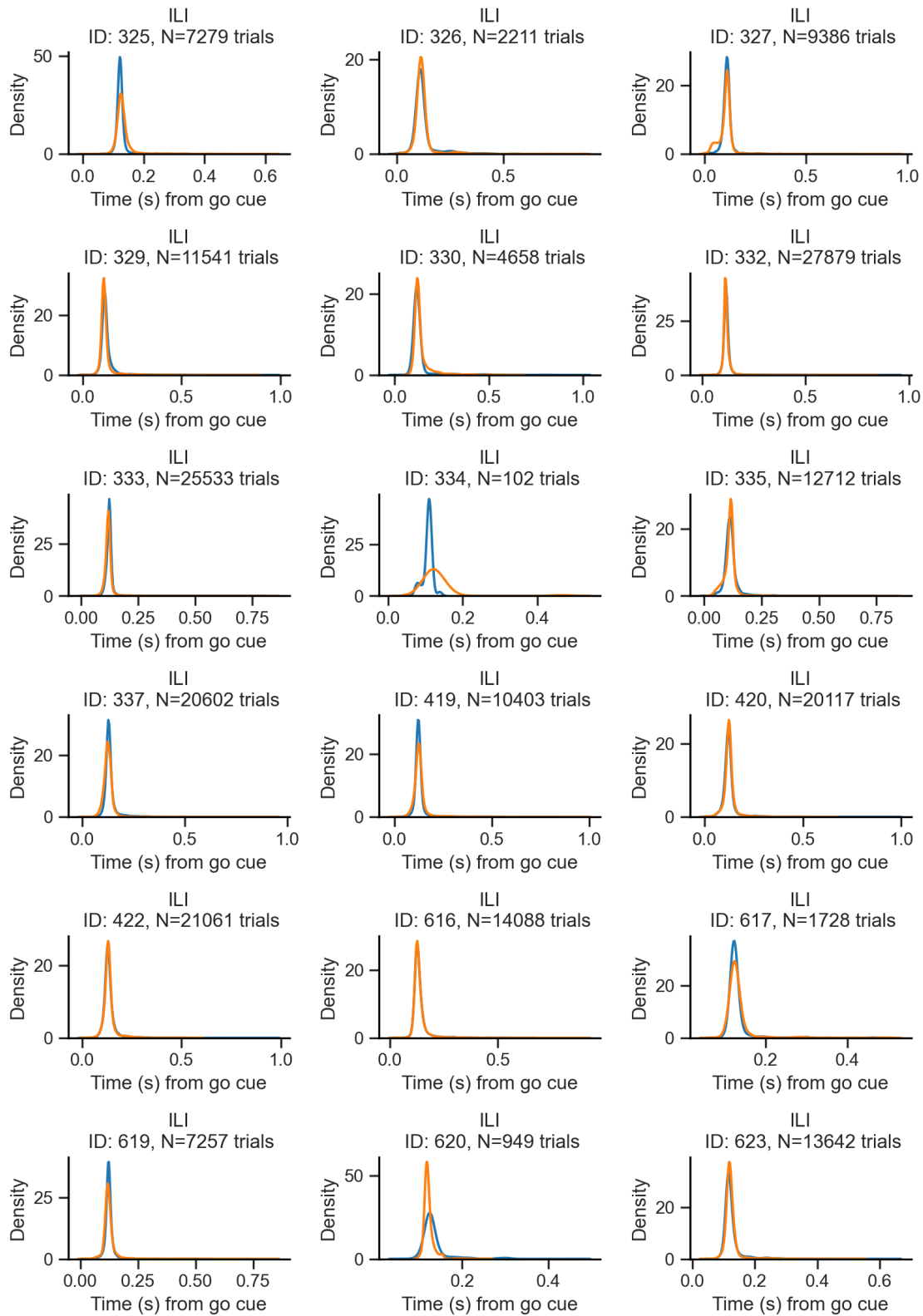
Repeat trial



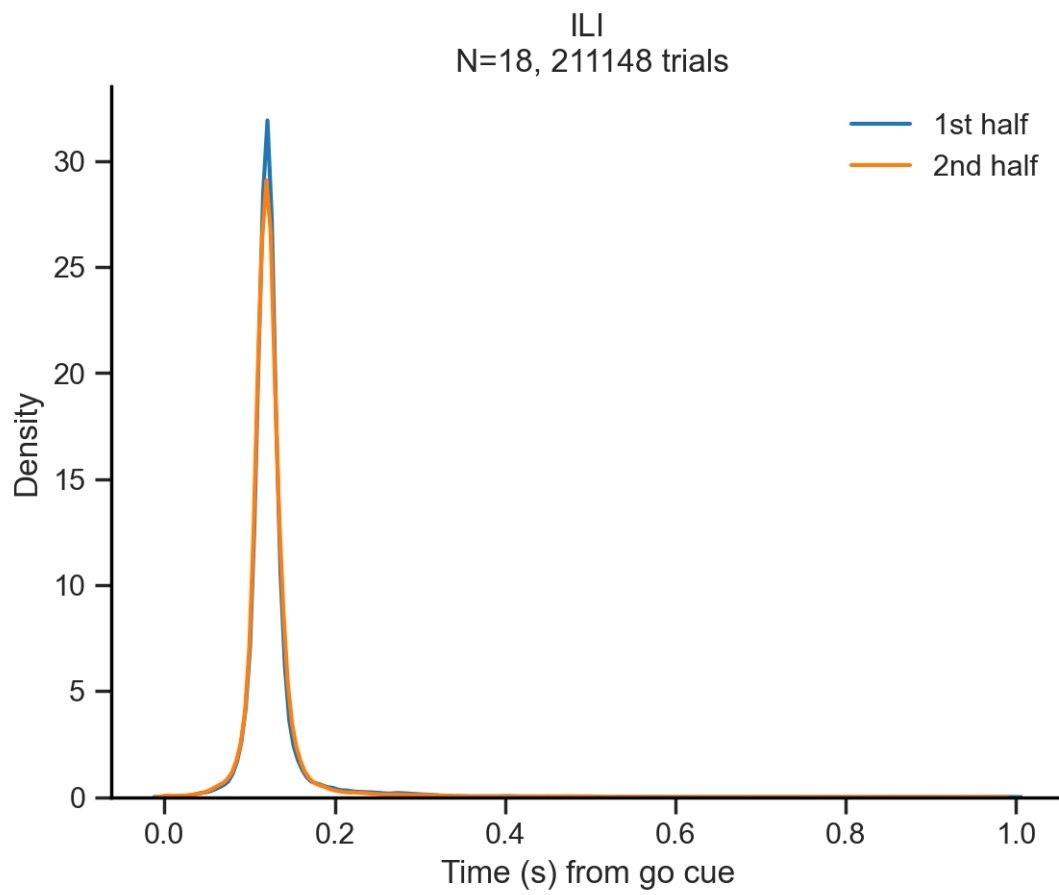


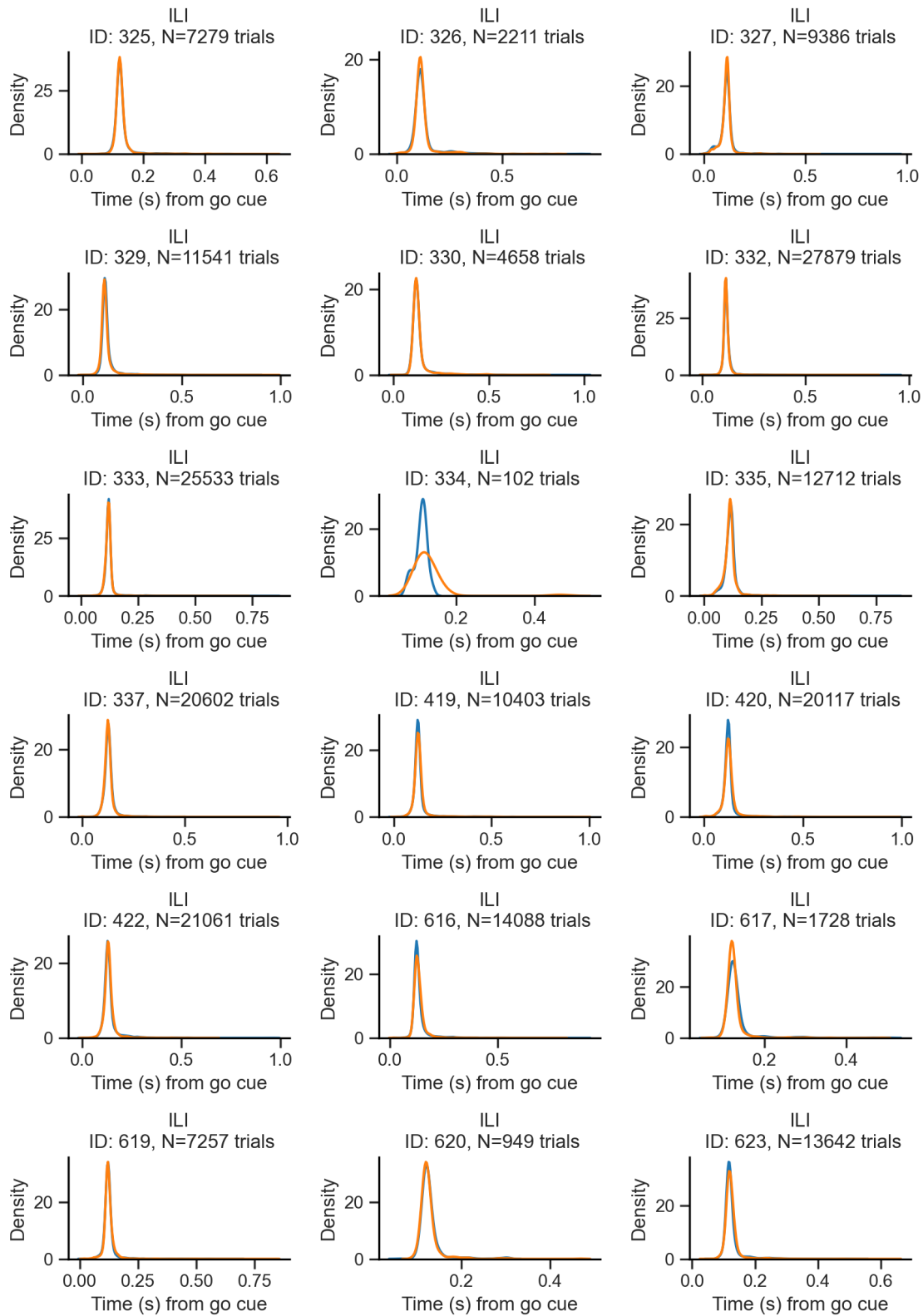
Stimulus





1st vs 2nd session half





Hidden Markov Model (HMM)

8 Save all notebook's figures

```
[NbConvertApp] Converting notebook notebook.ipynb to html
[NbConvertApp] WARNING | Alternative text is missing on 63 image(s).
[NbConvertApp] Writing 14190602 bytes to notebook.html

Saved 63 images to notebook_files/

[NbConvertApp] Converting notebook notebook.ipynb to pdf
[NbConvertApp] Support files will be in notebook_files\
[NbConvertApp] Making directory .\notebook_files
[NbConvertApp] Writing 48203 bytes to notebook.tex
[NbConvertApp] Building PDF
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']
[NbConvertApp] WARNING | b had problems, most likely because there were no
citations
[NbConvertApp] PDF successfully created
[NbConvertApp] Writing 9195255 bytes to notebook.pdf
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